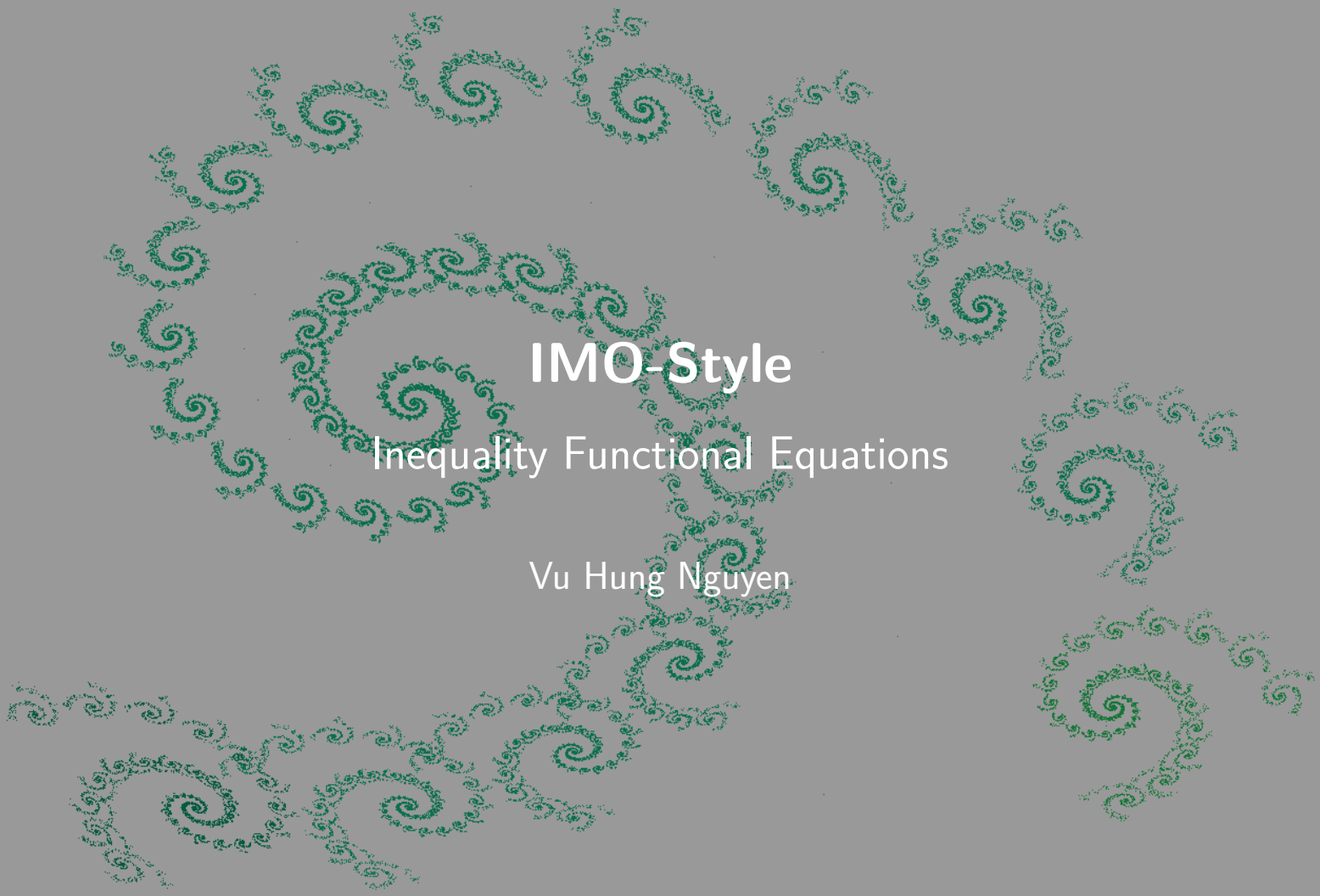


# IMO-Style

## Inequality Functional Equations

Vu Hung Nguyen

*Download PDF and resources:*

`https://github.com/vuhung16au/math-olympiad-ml`

*Read online at Vu's Maths Hub:*

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*"Exercise is the beste instrument in learnyng."*

Robert Recorde

# Table of Contents

**Preface** **5**

**Introduction** **5**

**Fundamental Review: Facts and Tools** **5**

**Frequently Seen Patterns** **6**

**Problems** **8**

Part I. Additive–Multiplicative Rigidity . . . . . 8
Problem 1. The Additive–Multiplicative Rigidity . . . . . 8
Problem 2. IMO 2013 – Problem 5 . . . . . 8
Problem 3. The Periodic Excess . . . . . 8
Problem 4. The Dyadic Lock . . . . . 9
Problem 5. The Triadic Balance . . . . . 9
Problem 34. The Ratio–Product Squeeze . . . . . 9
Problem 35. The Subadditive–Supermultiplicative Squeeze . . . . . 10
Part II. Reciprocal and Harmonic Transforms . . . . . 10
Problem 6. The Sum–Product Squeeze . . . . . 10
Problem 7. The Harmonic Squeeze . . . . . 10
Problem 8. The Multiplicative Mirror . . . . . 11
Problem 9. The Reflected Product . . . . . 11
Problem 10. The Reciprocal Pinch . . . . . 12
Problem 11. The Harmonic–Product Squeeze . . . . . 12
Part III. Iteration, Fixed Points, and Growth . . . . . 12
Problem 12. IMO 1977 – Problem 6 . . . . . 12
Problem 13. The Fixed-Point Lever . . . . . 13
Problem 14. IMO 2023 Shortlist – Algebra A4 . . . . . 13
Problem 15. FEOO 2020 Shortlist – Algebra A1 . . . . . 13
Problem 16. The Iterated Ratio Squeeze . . . . . 14
Problem 17. IMO 2022 Shortlist – Algebra A3 . . . . . 14
Problem 18. The Geometric-Mean Trap . . . . . 14
Problem 19. The Reciprocal-Slope Pinch . . . . . 15
Problem 20. The Cross-Dilated Shift . . . . . 15
Part IV. Mean and Norm Squeezes . . . . . 15
Problem 21. 2026 IMO Problem 5 . . . . . 15
Problem 22. The Hidden Quadratic Function . . . . . 16
Problem 23. The Euclidean–Multiplicative Squeeze . . . . . 16
Problem 24. The Quadratic Mean Squeeze . . . . . 16
Part V. Real-Line and Polynomial Inequalities . . . . . 17
Problem 25. Vietnam Mathematical Olympiad 1991 . . . . . 17
Problem 26. IMO 2023 Shortlist – Algebra A2 . . . . . 17
Problem 27. IMO 2011 – Problem 3 . . . . . 17
Problem 28. The Degree-Normalized Cubic Squeeze . . . . . 18
Problem 29. The Conditioned Cubic Squeeze . . . . . 18
Problem 30. The Quadratic–Difference Squeeze . . . . . 18
Problem 31. The Iterated Parallelogram–Quadratic Squeeze . . . . . 19
Problem 32. The Skew-Symmetric Parallelogram Squeeze . . . . . 19
Problem 33. The Dyadic Parallelogram–Quadratic Squeeze . . . . . 19

**Solutions** **20**
Part I. Additive–Multiplicative Rigidity . . . 20
Solution 1. The Additive–Multiplicative Rigidity . . . 20
Solution 2. IMO 2013 – Problem 5 . . . 21
Solution 3. The Periodic Excess . . . 21
Solution 4. The Dyadic Lock . . . 22
Solution 5. The Triadic Balance . . . 23
Solution 34. The Ratio–Product Squeeze . . . 24
Solution 35. The Subadditive–Supermultiplicative Squeeze . . . 25
Part II. Reciprocal and Harmonic Transforms . . . 27
Solution 6. The Sum–Product Squeeze . . . 27
Solution 7. The Harmonic Squeeze . . . 28
Solution 8. The Multiplicative Mirror . . . 29
Solution 9. The Reflected Product . . . 30
Solution 10. The Reciprocal Pinch . . . 31
Solution 11. The Harmonic–Product Squeeze . . . 32
Part III. Iteration, Fixed Points, and Growth . . . 33
Solution 12. IMO 1977 – Problem 6 . . . 33
Solution 13. The Fixed-Point Lever . . . 34
Solution 14. IMO 2023 Shortlist – Algebra A4 . . . 35
Solution 15. FEOO 2020 Shortlist – Algebra A1 . . . 36
Solution 16. The Iterated Ratio Squeeze . . . 37
Solution 17. IMO 2022 Shortlist – Algebra A3 . . . 38
Solution 18. The Geometric-Mean Trap . . . 41
Solution 19. The Reciprocal-Slope Pinch . . . 42
Solution 20. The Cross-Dilated Shift . . . 43
Part IV. Mean and Norm Squeezes . . . 44
Solution 21. 2026 IMO Problem 5 . . . 44
Solution 22. The Hidden Quadratic Function . . . 44
Solution 23. The Euclidean–Multiplicative Squeeze . . . 46
Solution 24. The Quadratic Mean Squeeze . . . 47
Part V. Real-Line and Polynomial Inequalities . . . 48
Solution 25. Vietnam Mathematical Olympiad 1991 . . . 48
Solution 26. IMO 2023 Shortlist – Algebra A2 . . . 49
Solution 27. IMO 2011 – Problem 3 . . . 50
Solution 28. The Degree-Normalized Cubic Squeeze . . . 51
Solution 29. The Conditioned Cubic Squeeze . . . 51
Solution 30. The Quadratic–Difference Squeeze . . . 53
Solution 31. The Iterated Parallelogram–Quadratic Squeeze . . . 53
Solution 32. The Skew-Symmetric Parallelogram Squeeze . . . 54
Solution 33. The Dyadic Parallelogram–Quadratic Squeeze . . . 56

**Conclusion** **57**

# Preface

This is the final edition of a booklet of olympiad-style functional equations and inequalities. It is written especially for students in Years 10–12 preparing for mathematical olympiads, as well as for anyone who enjoys practising with challenging, unusual, and unfamiliar problems. The aim is to develop flexible problem-solving habits, not just to collect answers.

# Introduction

The problems focus on olympiad-style functional equations built around inequalities. Their domains include the positive real numbers, the real numbers, the positive rationals, and the positive integers. The booklet is arranged in five themed parts: additive and multiplicative rigidity; reciprocal and harmonic transforms; iteration, fixed points, and growth; mean and norm squeezes; and real-line and polynomial inequalities. Problems come first, grouped by theme, and the full solutions appear together at the end.

For each problem, first read the statement carefully and try to solve it on your own. Pay attention to the domain and to what the inequalities say for special choices of inputs. If you get stuck, use the hint as a nudge rather than as a replacement for a proof. Once you have a solution, compare it with the one at the end: check each step, and note any idea that you did not use.

# Fundamental Review: Facts and Tools

The *squeeze principle* says that if a quantity is bounded above and below by the same value, it must equal that value. In these problems, one inequality often supplies an upper bound and another a matching lower bound, sometimes only after a substitution or iteration.

**Order and monotonicity.** Adding inequalities preserves their direction. Multiplying or dividing by a positive quantity also preserves the direction, while doing so by a negative quantity reverses it. Applying an increasing function preserves order; for positive numbers, taking reciprocals reverses order. A function is *nondecreasing* if $x \le y$ implies $f(x) \le f(y)$. For example, a positive superadditive function on the positive reals is strictly increasing: if $0 < x < y$, then $f(y) \ge f(x) + f(y-x) > f(x)$.

**Means.** For nonnegative $u, v$, the arithmetic–geometric mean inequality is

$$\frac{u+v}{2} \ge \sqrt{uv},$$

with equality exactly when $u = v$. The root-mean-square–arithmetic-mean inequality is

$$\sqrt{\frac{u^2+v^2}{2}} \ge \frac{u+v}{2} \qquad (u, v \ge 0),$$

again with equality exactly when $u = v$. These inequalities are useful when the problem places a function expression between two familiar means.

**Iteration and approximation.** Mathematical induction justifies repeated addition or multiplication, such as deriving bounds at integer multiples or powers. The Archimedean property says that the positive integers are unbounded. In particular, for $t > 0$,

$$0 \le t - \frac{\lfloor nt \rfloor}{n} < \frac{1}{n},$$

so $\lfloor nt \rfloor / n \to t$. Floor approximations of this kind let us compare a value at an arbitrary real input with values at integer multiples.

**Density and monotonicity.** The rational numbers are dense in the real numbers, and $\{2^q : q \in \mathbb{Q}\}$ is dense in the positive reals. If a nondecreasing function agrees with the identity on a dense subset of its domain, it agrees with the identity throughout: nearby points from the dense subset bound its value from below and above.

**Midpoints and quadratic identities.** The parallelogram identity

$$\frac{(x+y)^2 + (x-y)^2}{2} = x^2 + y^2$$

often simplifies inequalities on the real line. A related midpoint fact is that a monotone function on an interval satisfying

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$$

for all $x, y$ in that interval is affine. This is a regularity result for the midpoint Jensen equation; no continuity assumption needs to be added when monotonicity is known.

Basic algebra is used throughout: factor differences of squares, use the laws of exponents, and handle reciprocals carefully, noting that reciprocal order reversal requires positive quantities. Also choose inputs such as $x = y$, $y = 1$, or $y = -x$ only when the domain permits them. Normalizing a function, for example by setting $g(x) = f(x)/x^k$, can expose simpler behavior; always check that the normalization is defined on the stated domain.

# Frequently Seen Patterns

Look for special inputs that produce bounds, then combine or repeat those bounds until they meet. Swapping variables, using reciprocal or scaled inputs, and normalizing by a power often reveal hidden symmetry or monotonicity. Equality cases are especially informative: they can point to the candidate function and help show that no other function works.

# Table of Contents

# Problems

## Part I. Additive–Multiplicative Rigidity

### Problem 1. The Additive–Multiplicative Rigidity

Find all functions $f$ that map positive real numbers to positive real numbers and satisfy, for all $x, y > 0$,

$$f(x+y) \geq f(x)f(y) \geq f(xy).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Evaluate at 1, then compare repeated addition with repeated multiplication.

### Problem 2. IMO 2013 – Problem 5

Let $f$ be a function that maps positive rational numbers to real numbers and satisfies, for all positive rational numbers $x, y$,

$$f(x)f(y) \geq f(xy),$$
$$f(x+y) \geq f(x) + f(y).$$

Suppose there exists a rational number $a > 1$ such that $f(a) = a$.
Prove that

$$f(x) = x \quad \text{for all positive rational numbers } x.$$

*Source: International Mathematical Olympiad 2013 (54th IMO), Day 2, Problem 5. The wording has been slightly adapted without changing the mathematical content.*

**Hint:** Iterate both inequalities at sums and products of $a$; can these bounds trap values at other rationals?

### Problem 3. The Periodic Excess

Let $a > 1$ be a fixed real number. Find all functions $f$ that map positive real numbers to positive real numbers and satisfy, for all $x, y > 0$,

$$f(x+y) \geq f(x) + f(y),$$
$$f(xy) \leq f(x)f(y),$$
$$f(x+a) \leq f(x) + a.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Iterate the shift by $a$ and compare it with what repeated addition tells you.

### Problem 4. The Dyadic Lock

Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f(2)f(f(2)) \leq 4$$

and, for all $x, y > 0$,

$$f(x + y) \geq f(x) + f(y) \geq f(xy)\left(\frac{1}{f(x)} + \frac{1}{f(y)}\right).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** What does the inequality at equal inputs tell you, and how might it interact with the bound at 2?

### Problem 5. The Triadic Balance

Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f(3) + f\left(\frac{1}{3}\right) \leq \frac{10}{3},$$

and, for all $x, y > 0$,

$$f(x + y) \geq f(x) + f(y), \qquad f(xy) \leq f(x)f(y).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Use repeated addition and multiplication at 3 and $1/3$ to exploit the normalization bound.

### Problem 34. The Ratio–Product Squeeze

Find all functions $f : (0, \infty) \to (0, \infty)$ such that, for all $x, y > 0$,

$$xf(x + y) \geq (x + y)f(x), \qquad f(xy) \leq f(x)f(y),$$

and for which there exists $a > 1$ satisfying

$$f(f(a)) \leq f(a).$$

*Original problem for this collection; not an official IMO problem.*

**Hint:** Normalize by $x$ to reveal monotonicity and submultiplicativity, then propagate equality from an interval around 1 using integer roots.

### Problem 35. The Subadditive–Supermultiplicative Squeeze

Find all functions $f$ that map positive real numbers to positive real numbers, satisfy $f(f(a)) = a$ for some $a > 1$, and, for all $x, y > 0$,

$$\boxed{f(x) \le f(x+y) \le f(x) + f(y) \le f(xy)\left(\frac{1}{f(x)} + \frac{1}{f(y)}\right).}$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Extract monotonicity, subadditivity, and supermultiplicativity; then use the anchor to squeeze $f(x)$ against $x$.

## Part II. Reciprocal and Harmonic Transforms

### Problem 6. The Sum–Product Squeeze

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers and satisfy, for all $x, y > 0$,

$$f(x+y)\big(f(x) + f(y)\big) \le 2f(2)f(x)f(y) \le f(xy).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** What do the inequalities become for carefully chosen pairs such as $(x, 1)$ or $(x, x)$?

### Problem 7. The Harmonic Squeeze

Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f(2) \ge \frac{1}{2}$$

and, for all $x, y > 0$,

$$f(x+y)\big(f(x) + f(y)\big) \le f(x)f(y) \le f(xy).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Set $x = y$ first, then compare the resulting doubling relation with the product bound.

### Problem 8. The Multiplicative Mirror

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers such that

$$f(x)f\left(\frac{1}{x}\right) = 1, \qquad f(f(2)) \le 2,$$

and, for all $x, y > 0$,

$$\frac{f(x)f(y)}{f(x+y)\big(f(x)+f(y)\big)} \ge 1 \ge \frac{f(x)^2}{f(xy)f(x/y)}.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** What do the two inequalities say after replacing both variables by their reciprocals?

### Problem 9. The Reflected Product

Find all functions $f : (0,\infty) \to (0,\infty)$ such that

$$f(x)f\left(\frac{1}{x}\right) = 1, \qquad f(2) \le 2,$$

and, for all $x, y > 0$,

$$f(x+y) \ge f(x) + f(y), \qquad f(x)^2 \le f(xy)f\left(\frac{x}{y}\right).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Combine the reciprocal identity with the product inequality after interchanging $x$ and $y$.

### Problem 10. The Reciprocal Pinch

Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f(x)f\left(\frac{1}{x}\right) = 1, \qquad f(2) + f\left(\frac{1}{2}\right) \leq \frac{5}{2},$$

and, for all $x, y > 0$,

$$f(x + y) \geq f(x) + f(y), \qquad f(xy) \geq f(x)f(y).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Use reciprocal symmetry at 2 and compare the resulting values with their given sum bound.

### Problem 11. The Harmonic–Product Squeeze

Find all functions $f$ that map positive real numbers to positive real numbers, provided that $f(f(c)) \geq c$ for some $c \in (0, 1)$, and, for all $x, y > 0$,

$$f\left(\frac{xy}{x + y}\right) \leq \frac{f(x)f(y)}{f(x) + f(y)}, \qquad f(xy) \geq f(x)f(y).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Try reciprocal substitutions, then bring the special point $c$ into play.

## Part III. Iteration, Fixed Points, and Growth

### Problem 12. IMO 1977 – Problem 6

Let $f$ be a function that maps positive integers to positive integers and satisfies, for all $n \in \mathbb{Z}_{>0}$,

$$f(n + 1) > f(f(n)).$$

Prove that

$$f(n) = n \qquad (n \in \mathbb{Z}_{>0}).$$

*Source: International Mathematical Olympiad 1977 (19th IMO), Day 2, Problem 6.*

**Hint:** Can you compare consecutive values of $f$ by combining the condition with iteration?

### Problem 13. The Fixed-Point Lever

Find all functions $f$ that map positive real numbers to positive real numbers, if there exists $a > 0$ such that $f(f(a)) = a$, and for all $x, y > 0$,

$$f(x)\big(x + f(y)\big) \geq f(f(x))\big(f(x) + y\big).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Try using the special point $a$ in each variable, where $f(f(a)) = a$.

### Problem 14. IMO 2023 Shortlist – Algebra A4

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers such that, for all $x, y > 0$,

$$x\big(f(x) + f(y)\big) \geq \big(f(f(x)) + y\big)f(y).$$

*Source: This problem is taken from the IMO 2023 Shortlist, Algebra Problem A4. The wording has been slightly adapted without changing the mathematical content.*

**Hint:** Fix $x$ and let $y$ vary; what does the inequality force about the possible size of $f(y)$?

### Problem 15. FEOO 2020 Shortlist – Algebra A1

Find all functions $f$ of a single variable that map real numbers to real numbers and satisfy, for all $x, y \in \mathbb{R}$,

$$2xyf(xy) \geq x^2y^2 + xf(x)f(y^2).$$

*Source: Functional Equations Online Olympiad (FEOO) 2020, Shortlist, Algebra Problem A1. The wording has been slightly adapted without changing the mathematical content.*

**Hint:** Start with $y = 0$ and exploit the fact that $y^2$ stays nonnegative.

### Problem 16. The Iterated Ratio Squeeze

Let $a > 1$ be a fixed real number. Find all functions $f$ that map positive real numbers to positive real numbers, satisfy $f(a) = a$, and, for all $x, y > 0$,

$$\frac{f(x)}{f(y)} + \frac{f(y)}{f(x)} \geq \frac{x}{y} + \frac{y}{x} \geq \frac{f^{\circ 3}(x)}{f^{\circ 3}(y)} + \frac{f^{\circ 3}(y)}{f^{\circ 3}(x)}.$$

Here $f^{\circ 3}(x) = f(f(f(x)))$.

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Fix one variable at $a$ and follow what the squeeze says through repeated applications of $f$.

### Problem 17. IMO 2022 Shortlist – Algebra A3

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers such that, for every $x > 0$, there exists exactly one $y > 0$ satisfying

$$xf(y) + yf(x) \leq 2.$$

*Source: International Mathematical Olympiad 2022 (63rd IMO), Shortlist, Algebra Problem A3, proposed by the Netherlands. The wording has been slightly adapted without changing the mathematical content.*

**Hint:** For a fixed $x$, what shape does the set of admissible $y$ have, and how does uniqueness restrict it?

### Problem 18. The Geometric-Mean Trap

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers and satisfy the following property: for every $x > 0$, there exists exactly one $y > 0$ such that

$$\frac{x}{f(y)} + \frac{y}{f(x)} \leq \frac{f(x)}{x} + \frac{f(y)}{y}.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** For fixed $x$, rewrite the condition using the ratios $f(x)/x$ and $f(y)/y$; what does uniqueness imply?

### Problem 19. The Reciprocal-Slope Pinch

Find all functions $f : (0, \infty) \to (0, \infty)$ such that, for all $x, y > 0$,

$$xy + f\big(xf(x)\big) \le xf(x + y).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** For fixed $x$, compare the lower bounds at nearby inputs and watch how the term $xf(x)$ enters.

### Problem 20. The Cross-Dilated Shift

Find all functions $f : (0, \infty) \to (0, \infty)$ such that, for all $x, y > 0$,

$$x^2 + f\big(xf(y)\big) \le xf(x + y).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Compare the inequality at exchanged or rescaled inputs; what role does the sum $x + y$ play?

## Part IV. Mean and Norm Squeezes

### Problem 21. 2026 IMO Problem 5

Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that, for all $x, y > 0$,

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \ge \frac{f(x) + y}{2} \ge \sqrt{xf(y)}.$$

*Source: 2026 IMO, Problem 5.*

**Hint:** What can you learn by choosing $y$ in relation to $f(x)$ and comparing the two mean bounds?

### Problem 22. The Hidden Quadratic Function

Find all functions $f$ of a single variable that map positive real numbers to positive real numbers such that $f(f(2)) \leq 16$ and, for all $x, y > 0$,

$$\sqrt{f(x+y)} \geq \sqrt{f(x)} + \sqrt{f(y)} \geq \sqrt{f(xy)} \left( \frac{1}{\sqrt{f(x)}} + \frac{1}{\sqrt{f(y)}} \right).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Try setting $x = y$ and compare the resulting bounds as the input is doubled.

### Problem 23. The Euclidean–Multiplicative Squeeze

Find all functions $f$ that map positive real numbers to positive real numbers, satisfy

$$f(f(\sqrt{2})) \leq \sqrt{2},$$

and, for all $x, y > 0$,

$$f\left(\sqrt{x^2+y^2}\right)^2 \geq f(x)^2 + f(y)^2 \geq f(xy)^2 \left( \frac{1}{f(x)^2} + \frac{1}{f(y)^2} \right).$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Can you choose inputs so that the Euclidean combination and the product are closely related?

### Problem 24. The Quadratic Mean Squeeze

Find all functions $f$ that map positive real numbers to positive real numbers, satisfy $f(f(a)) \leq a^4$ for some $a > 0$, and, for all $x, y > 0$,

$$\sqrt{\frac{x^4 + f(y)^2}{2}} \geq \frac{f(x) + y^2}{2} \geq x\sqrt{f(y)}.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Look for choices of $x$ and $y$ that make the two mean bounds especially tight.

# Part V. Real-Line and Polynomial Inequalities

### Problem 25. Vietnam Mathematical Olympiad 1991

Find all functions $f$ of a single variable that map real numbers to real numbers and satisfy, for all real numbers $x, y, z$,

$$\frac{f(xy) + f(xz)}{2} - f(x)f(yz) \geq \frac{1}{4}.$$

*Source: 29th Vietnamese Mathematical Olympiad, 1991, First Day, Problem 1.*

**Hint:** What do the substitutions $y = z$ and $y = 0$ reveal?

### Problem 26. IMO 2023 Shortlist – Algebra A2

Let $f$ be a function that maps real numbers to real numbers such that, for all $x, y \in \mathbb{R}$,

$$f(x + y)f(x - y) \geq f(x)^2 - f(y)^2.$$

Suppose that the inequality is strict for at least one pair of real numbers $(x_0, y_0)$.
Prove that either

$$f(x) \geq 0 \quad \text{for all } x \in \mathbb{R}$$

or

$$f(x) \leq 0 \quad \text{for all } x \in \mathbb{R}.$$

*Source: IMO 2023 Shortlist, Algebra Problem A2, proposed by Malaysia. The wording has been slightly adapted without changing the mathematical content.*

**Hint:** Compare the inequality with the one obtained by swapping $x$ and $y$.

### Problem 27. IMO 2011 – Problem 3

Let $f$ be a function that maps real numbers to real numbers and satisfies, for all $x, y \in \mathbb{R}$,

$$f(x + y) \leq yf(x) + f(f(x)).$$

Prove that

$$f(x) = 0 \qquad \text{for all } x \leq 0.$$

*Source: International Mathematical Olympiad 2011 (52nd IMO), Day 1, Problem 3.*

**Hint:** Begin with $y = 0$ and $y = -x$; what do these say about the values at 0 and $f(x)$?

### Problem 28. The Degree-Normalized Cubic Squeeze

Find all functions $f$ that map positive real numbers to real numbers and satisfy, for all $x, y > 0$,

$$\frac{f(x)}{x^2} + \frac{f(y)}{y^2} \le \frac{f(x+y)}{(x+y)^2} \qquad \text{and} \qquad \frac{f(x)}{x^3} + \frac{f(y)}{y^3} \ge \frac{2f(x+y)}{(x+y)^3}.$$

**Hint:** Normalize by a power of the input; which choice makes one of the inequalities simpler?

### Problem 29. The Conditioned Cubic Squeeze

Find all functions $f$ that map real numbers to real numbers such that $f(0) = 0$, $f(1) = 1$, and, for all $a, b, c \in \mathbb{R}$:

1. If $a + b + c \ge 0$, then $f(a^3) + f(b^3) + f(c^3) \ge 3f(abc)$.
2. If $a + b + c \le 0$, then $f(a^3) + f(b^3) + f(c^3) \le 3f(abc)$.

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Try triples with a zero entry or two equal entries, and compare the cases of opposite signs.

### Problem 30. The Quadratic–Difference Squeeze

Find all functions $f$ that map real numbers to real numbers and satisfy, for all $x, y \in \mathbb{R}$,

$$f(x) + f(y) \ge 2xy \qquad \text{and} \qquad f(x+y) + f(x-y) \le 2f(x) + 2y^2.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Use the first inequality to bound selected values, then feed those bounds into the second.

### Problem 31. The Iterated Parallelogram–Quadratic Squeeze

Find all functions $f$ that map real numbers to real numbers, satisfy

$$f(f(2)) \le 16,$$

and, for all $x, y \in \mathbb{R}$,

$$x^2 - y^2 + 2yf(x) + f(y) - f(f(x)) \le \frac{f(x+y) + f(x-y)}{2} \le f(x) + y^2.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Start with $x = 0$ and $y = 0$; what bounds do they give before you iterate $f$?

### Problem 32. The Skew-Symmetric Parallelogram Squeeze

Find all functions $f$ that map real numbers to real numbers and satisfy, for all $x, y \in \mathbb{R}$,

$$2xy + x^2 f(y) - y^2 f(x) \le \frac{f(x+y) + f(x-y)}{2} \le f(x) + y^2.$$

*Composed by Vu Hung Nguyen, 2026*

**Hint:** Swap $x$ and $y$ in the lower bound and compare the result with the upper bound.

### Problem 33. The Dyadic Parallelogram–Quadratic Squeeze

Find all functions $f$ that map real numbers to real numbers, satisfy

$$f(f(1)) \le 1,$$

and, for all $x, y \in \mathbb{R}$,

$$8xy + 3f(x) - 3f(y) \le \frac{5\big(f(x+y) + f(x-y)\big)}{2} \le 5f(x) + 5y^2.$$

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**Hint:** First set one variable to zero; can you extract a simpler relation before using the iterate bound?

# Solutions

## Part I. Additive–Multiplicative Rigidity

### Solution 1. The Additive–Multiplicative Rigidity

*Proof.* Set $g(x) = \ln f(x)$. Since ln is strictly increasing, the given inequalities are equivalent to

$$g(x+y) \geq g(x) + g(y), \qquad g(xy) \leq g(x) + g(y). \tag{1}$$

**Step 1. Show that $g$ vanishes on the positive integers.**
Taking $x = y = 1$ in the second inequality of (1) gives $g(1) \geq 0$. Repeated superadditivity then gives

$$g(n) \geq ng(1) \geq 0 \qquad (n \in \mathbb{N}).$$

Fix an integer $n \geq 2$. For every positive integer $m$, repeated application of the two inequalities in (1) gives

$$n^{m-1}g(n) \leq g(n^m) \leq mg(n).$$

If $g(n) > 0$, division by $g(n)$ would imply $n^{m-1} \leq m$ for every positive integer $m$, which is impossible for sufficiently large $m$. Since $g(n) \geq 0$, it follows that $g(n) = 0$ for each integer $n \geq 2$.
Superadditivity gives $g(2) \geq 2g(1)$. As $g(2) = 0$ and $g(1) \geq 0$, we obtain $g(1) = 0$. Thus

$$g(n) = 0 \qquad (n \in \mathbb{N}). \tag{2}$$

**Step 2. Show that $g$ vanishes on positive rationals.**
Write a positive rational as $r = p/q$, where $p, q$ are positive integers. By superadditivity and (2),

$$0 = g(p) = g(\underbrace{r + \cdots + r}_{q \text{ terms}}) \geq qg(r),$$

so $g(r) \leq 0$. By the second inequality in (1) and (2),

$$0 = g(p) = g(rq) \leq g(r) + g(q) = g(r),$$

so $g(r) \geq 0$. Therefore

$$g(r) = 0 \qquad (r \in \mathbb{Q}_{>0}). \tag{3}$$

**Step 3. Show that $g$ vanishes everywhere.**
Fix $x > 0$ and an integer $n \geq 2$. Using $g(n) = 0$ in (1),

$$g(x) = g\left(\frac{x}{n} \cdot n\right) \leq g\left(\frac{x}{n}\right),$$

while superadditivity gives

$$g(x) \geq ng\left(\frac{x}{n}\right).$$

Hence

$$g(x) \leq g\left(\frac{x}{n}\right) \leq \frac{g(x)}{n}.$$

If $g(x) > 0$, these bounds contradict $n \geq 2$. Thus $g(x) \leq 0$ for every $x > 0$.

Finally, by (1) and (2),

$$0 = g(1) = g\left(x \cdot \frac{1}{x}\right) \leq g(x) + g\left(\frac{1}{x}\right).$$

Both terms on the right are nonpositive, so each must be zero. In particular, $g(x) = 0$ for all $x > 0$. Therefore $f(x) = e^{g(x)} = 1$.

**Verification.**
The constant function $f(x) = 1$ makes both inequalities equalities, so it is indeed the unique solution. □

## Solution 2. IMO 2013 – Problem 5

*Proof.* From $f(a)f(1) \geq f(a)$, we have $f(1) \geq 1$, hence $f(n) \geq n$ for positive integers $n$. Also, $f(q)f(p/q) \geq f(p) > 0$ shows $f(r) > 0$ for every positive rational $r$. Thus superadditivity makes $f$ strictly increasing.
Submultiplicativity gives $f(a^n) \leq a^n$, while $f(a^n)f(a^{-n}) \geq f(1) \geq 1$ yields $f(a^{-n}) \geq a^{-n}$. For rational $x > 0$, set $m_n = \lfloor a^n x \rfloor$. For large $n$, monotonicity and superadditivity give

$$f(x) \geq f(m_n a^{-n}) \geq m_n f(a^{-n}) \geq \frac{\lfloor a^n x \rfloor}{a^n} \longrightarrow x.$$

Hence $f(x) \geq x$.
Applying this to $a^n$ shows $f(a^n) = a^n$. For any positive rational $x$, choose $n$ with $a^n > x$. Then

$$a^n = f(a^n) \geq f(x) + f(a^n - x) \geq f(x) + a^n - x,$$

so $f(x) \leq x$. Therefore $f(x) = x$ for all positive rationals; the identity satisfies both inequalities and the fixed-point condition. □

## Solution 3. The Periodic Excess

*Proof.* **Step 1. Establish monotonicity and a growth bound.**
Since $f$ is positive and superadditive, it is strictly increasing: if $y > x > 0$, then

$$f(y) = f(x + (y - x)) \geq f(x) + f(y - x) > f(x).$$

Taking $x = y = 1$ in the multiplicative inequality gives $f(1) \leq f(1)^2$, so $f(1) \geq 1$. Repeated superadditivity then implies

$$f(n) \geq n \qquad (n \in \mathbb{N}).$$

Fix $t > 1$ and a positive integer $n$. Repeated submultiplicativity and monotonicity give

$$f(t)^n \geq f(t^n) \geq f(\lfloor t^n \rfloor) \geq \lfloor t^n \rfloor.$$

Taking $n$th roots and letting $n \to \infty$ yields

$$f(t) \geq t \qquad (t > 1). \tag{1}$$

**Step 2. Force an exact translation law.**
Superadditivity and the third condition imply

$$f(x) + f(a) \le f(x + a) \le f(x) + a.$$

Thus $f(a) \le a$. Since $a > 1$, (1) also gives $f(a) \ge a$. Therefore

$$f(a) = a, \qquad f(x + a) = f(x) + a \quad (x > 0). \tag{2}$$

**Step 3. Study the periodic excess.**
Define $h(x) = f(x) - x$. Equation (2) gives

$$h(x + a) = h(x),$$

so $h$ is periodic with period $a$ on the positive reals. Superadditivity of $f$ gives

$$h(x + y) \ge h(x) + h(y). \tag{3}$$

For any $x > 0$, choose a positive integer $k$ large enough that $x + ka > 1$. By periodicity and (1),

$$h(x) = h(x + ka) = f(x + ka) - (x + ka) \ge 0.$$

Since $f$ is increasing and $f(a) = a$, for $0 < x \le a$ we have

$$h(x) = f(x) - x \le a - x < a.$$

By periodicity, this yields

$$0 \le h(x) \le a \qquad (x > 0). \tag{4}$$

**Step 4. Show that the periodic excess vanishes.**
Repeated use of (3) gives $h(nx) \ge nh(x)$ for every positive integer $n$. By (4),

$$0 \le nh(x) \le h(nx) \le a.$$

Hence $0 \le h(x) \le a/n$ for every positive integer $n$, forcing $h(x) = 0$. Thus $f(x) = x$ for all $x > 0$.

**Verification.**
The identity function satisfies all three inequalities with equality, so it is the unique solution. □

## Solution 4. The Dyadic Lock

*Proof.* The second inequality in the problem implies

$$f(xy) \le f(x)f(y). \tag{1}$$

Also, $f$ is strictly increasing: if $u > v > 0$, then $f(u) = f(v + (u - v)) \ge f(v) + f(u - v) > f(v)$. Taking $x = y = 1$ in (1) gives $f(1) \ge 1$. Superadditivity then gives $f(2) \ge 2f(1) \ge 2$. Since $f$ is increasing, $f(f(2)) \ge f(2)$, and therefore

$$4 \ge f(2)f(f(2)) \ge f(2)^2 \ge 4.$$

It follows that $f(2) = 2$ and, from $f(2) \geq 2f(1) \geq 2$, also $f(1) = 1$. Superadditivity and (1), applied repeatedly, now give

$$f(2^n) = 2^n \qquad (n = 0, 1, 2, \ldots). \tag{2}$$

We next obtain a common upper bound. Fix $t > 0$, and for all sufficiently large $n$ let

$$m_n = \left\lfloor \frac{2^n}{t} \right\rfloor.$$

By monotonicity, superadditivity and (2),

$$2^n = f(2^n) \geq f(m_n t) \geq m_n f(t).$$

Since $2^n/m_n \to t$, this proves

$$f(t) \leq t \qquad (t > 0). \tag{3}$$

For every positive integer $k$, superadditivity and (3) give

$$k = kf(1) \leq f(k) \leq k,$$

so $f(k) = k$. If $r = p/q$ is a positive rational, where $p, q$ are positive integers, then (1) yields

$$p = f(p) = f(qr) \leq f(q)f(r) = qf(r).$$

Hence $f(r) \geq r$, and (3) gives $f(r) = r$. For any positive real $x$, choose positive rationals $r < x < s$ arbitrarily close to $x$. Monotonicity then gives $r = f(r) \leq f(x) \leq f(s) = s$. Letting $r, s$ approach $x$ proves $f(x) = x$.
The identity function satisfies the original inequalities and numerical condition with equality, so it is the unique solution. □

## Solution 5. The Triadic Balance

*Proof.* **Step 1. Unlock the value at 3.**
Taking $x = y = 1$ in submultiplicativity gives

$$f(1) \leq f(1)^2,$$

so $f(1) \geq 1$. By superadditivity,

$$f(3) \geq 3f(1) \geq 3.$$

Also, submultiplicativity gives

$$f(1) = f\left(3 \cdot \frac{1}{3}\right) \leq f(3) f\left(\frac{1}{3}\right),$$

and hence

$$f\left(\frac{1}{3}\right) \geq \frac{f(1)}{f(3)} \geq \frac{1}{f(3)}.$$

Writing $a = f(3) \geq 3$, we obtain

$$\frac{10}{3} \geq f(3) + f\left(\frac{1}{3}\right) \geq a + \frac{1}{a} \geq 3 + \frac{1}{3} = \frac{10}{3}.$$

Equality holds throughout, so

$$f(1) = 1, \qquad f(3) = 3, \qquad f\left(\frac{1}{3}\right) = \frac{1}{3}.$$

**Step 2. Establish an upper bound.**
By induction, superadditivity and submultiplicativity give

$$f(3^n) = 3^n \qquad (n = 0, 1, 2, \ldots).$$

The function is strictly increasing: if $u > v > 0$, then

$$f(u) = f\big(v + (u - v)\big) \geq f(v) + f(u - v) > f(v).$$

Fix $t > 0$, and for all sufficiently large $n$ let

$$m_n = \left\lfloor \frac{3^n}{t} \right\rfloor.$$

Since $m_n t \leq 3^n$, monotonicity and superadditivity yield

$$3^n = f(3^n) \geq f(m_n t) \geq m_n f(t).$$

As $3^n/m_n \to t$, it follows that

$$f(t) \leq t \qquad (t > 0). \tag{1}$$

**Method I. Rational squeeze.**
For every positive integer $k$, superadditivity and (1) give

$$k = kf(1) \leq f(k) \leq k,$$

so $f(k) = k$. If $r = p/q$ is a positive rational, where $p, q$ are positive integers, submultiplicativity gives

$$p = f(p) = f(qr) \leq f(q)f(r) = qf(r).$$

Thus $f(r) \geq r$, and (1) shows $f(r) = r$. By monotonicity and the density of the positive rationals, $f(x) = x$ for all $x > 0$.

□

The identity function satisfies the original conditions with equality throughout, so it is the unique solution.

## Solution 34. The Ratio–Product Squeeze

*Proof.* Define

$$g(x) = \frac{f(x)}{x} \qquad (x > 0).$$

The first inequality implies

$$g(x + y) \geq g(x),$$

so $g$ is nondecreasing. The second inequality becomes

$$g(xy) \le g(x)g(y). \tag{1}$$

Taking $x = y = 1$ in (1) gives $g(1) \le g(1)^2$. Since $g(1) > 0$,

$$g(1) \ge 1. \tag{2}$$

Let $a > 1$ satisfy the composition condition. By monotonicity and (2),

$$g(a) \ge g(1) \ge 1, \qquad f(a) = ag(a) \ge a > 1.$$

Therefore $g(f(a)) \ge g(a) \ge 1$. On the other hand,

$$f(f(a)) = f(a)g(f(a)) \le f(a),$$

so $g(f(a)) \le 1$. It follows that

$$g(f(a)) = g(a) = g(1) = 1.$$

By monotonicity,

$$g(x) = 1 \qquad (1 \le x \le a). \tag{3}$$

Now let $1/a \le x \le 1$. From monotonicity and (3), $g(x) \le 1$ and $g(1/x) = 1$. Applying (1) to $x$ and $1/x$ gives

$$1 = g(1) \le g(x)g(1/x) = g(x),$$

so $g(x) = 1$. Hence

$$g(x) = 1 \qquad (1/a \le x \le a). \tag{4}$$

Finally, fix any $x > 0$. Choose a positive integer $n$ large enough that $t = x^{1/n} \in [1/a, a]$. By (4), $g(t) = g(1/t) = 1$. Repeatedly applying (1) yields

$$g(x) = g(t^n) \le g(t)^n = 1, \qquad g(1/x) = g((1/t)^n) \le g(1/t)^n = 1.$$

But (1) also gives

$$1 = g(1) \le g(x)g(1/x).$$

Both factors are positive and at most 1, so they are both equal to 1. In particular, $g(x) = 1$. Thus $g \equiv 1$, and consequently

$$f(x) = x \qquad (x > 0).$$

□

The identity function satisfies both inequalities with equality and $f(f(a)) = f(a)$, so it is the unique solution.

## Solution 35. The Subadditive–Supermultiplicative Squeeze

*Proof.* The first inequality shows that $f$ is nondecreasing, and the second shows that $f$

is subadditive. Since $f(x) + f(y) > 0$, the last inequality is equivalent to

$$f(xy) \geq f(x)f(y). \tag{1}$$

Thus $f$ is also supermultiplicative.

**Step 1. The anchor is a fixed point.**
Let $b = f(a)$. If $b > a$, monotonicity gives

$$a = f(b) \geq f(a) = b > a,$$

a contradiction. If $b < a$, then

$$a = f(b) \leq f(a) = b < a,$$

again a contradiction. Hence

$$f(a) = a. \tag{2}$$

**Step 2. Determine $f(1)$.**
Set $c = f(1)$. By (1), $c \geq c^2$, so $c \leq 1$. For every positive integer $n$, repeated use of (1) and (2) gives

$$f(a^n) \geq f(a)^n = a^n.$$

On the other hand, monotonicity and subadditivity give

$$f(a^n) \leq f(\lceil a^n \rceil) \leq \lceil a^n \rceil c.$$

Therefore

$$c \geq \frac{a^n}{\lceil a^n \rceil} \longrightarrow 1,$$

and so

$$f(1) = 1. \tag{3}$$

**Step 3. Show that $f(x) \geq x$.**
Fix $x > 0$. For all sufficiently large $n$, set

$$m_n = \left\lfloor \frac{a^n}{x} \right\rfloor, \qquad r_n = a^n - m_n x,$$

so $m_n \geq 1$ and $0 \leq r_n < x$. By (1) and (2),

$$a^n \leq f(a^n).$$

If $r_n > 0$, subadditivity, repeated subadditivity, and monotonicity give

$$f(a^n) = f(m_n x + r_n) \leq f(m_n x) + f(r_n) \leq m_n f(x) + f(x).$$

If $r_n = 0$, the same final bound follows directly from $f(m_n x) \leq m_n f(x)$. Thus

$$f(x) \geq \frac{a^n}{m_n + 1} = \frac{a^n}{\lfloor a^n/x \rfloor + 1}.$$

Letting $n \to \infty$ yields

$$f(x) \geq x. \tag{4}$$

**Step 4. Force equality.**
Using (1), (3), and (4) applied to $1/x$, we obtain

$$1 = f(1) = f\left(x \cdot \frac{1}{x}\right) \geq f(x) f\left(\frac{1}{x}\right) \geq \frac{f(x)}{x}.$$

Hence $f(x) \leq x$. Together with (4), this proves

$$f(x) = x \qquad (x > 0).$$

The identity function satisfies the chain and $f(f(a)) = a$ for every $a > 1$, so it is the unique solution. □

# Part II. Reciprocal and Harmonic Transforms

## Solution 6. The Sum–Product Squeeze

*Proof.* Set

$$g(x) = \frac{1}{f(x)}, \qquad k = \frac{g(2)}{2}.$$

The given inequalities are equivalent to

$$g(x+y) \geq k(g(x) + g(y)), \tag{1}$$
$$g(xy) \leq k\, g(x)g(y). \tag{2}$$

**Step 1.** Setting $x = y = 1$ gives

$$g(1) \leq 1, \qquad kg(1) \geq 1,$$

so $k \geq 1$.
If $k > 1$, repeated application of (1), with increments $1/n$, yields

$$g(2) \geq k^n g(1).$$

Letting $n \to \infty$ gives a contradiction. Thus $k = 1$, and consequently

$$g(1) = 1, \qquad g(2) = 2.$$

**Step 2.** By (1) and (2),

$$2g(x) \leq g(2x) \leq g(2)g(x) = 2g(x).$$

Hence

$$\boxed{g(2x) = 2g(x).}$$

In particular, $g(2^n) = 2^n$ for every positive integer $n$.
**Step 3.** As $g$ is positive and superadditive, it is strictly increasing.
Fix $x > 0$ and put

$$m_n = \left\lfloor \frac{2^n}{x} \right\rfloor.$$

For all sufficiently large $n$,

$$2^n = g(2^n) \geq g(m_n x) \geq m_n g(x).$$

Therefore,

$$g(x) \leq \frac{2^n}{\lfloor 2^n/x \rfloor} \xrightarrow[n\to\infty]{} x.$$

Thus $g(x) \leq x$ for every $x > 0$.
Conversely, by (2),

$$1 = g(1) \leq g(x)g(1/x) \leq \frac{g(x)}{x},$$

so $g(x) \geq x$.
We conclude that $g(x) = x$, whence

$$\boxed{f(x) = \frac{1}{x}}.$$

Finally, direct substitution gives

$$f(x+y)(f(x)+f(y)) = 2f(2)f(x)f(y) = f(xy) = \frac{1}{xy}.$$

Thus the proposed function satisfies both inequalities. □

## Solution 7. The Harmonic Squeeze

*Proof.* Set

$$g(x) = \frac{1}{f(x)}.$$

Since all function values are positive, taking reciprocals in the given inequalities gives

$$g(x+y) \geq g(x) + g(y), \qquad g(xy) \leq g(x)g(y),$$

and the condition on $f(2)$ gives $g(2) \leq 2$.

**Step 1. Lock the dyadic values.**
Taking $x = y = 1$ in submultiplicativity yields

$$g(1) \leq g(1)^2,$$

so $g(1) \geq 1$. Therefore

$$2 \geq g(2) \geq 2g(1) \geq 2,$$

which implies $g(1) = 1$ and $g(2) = 2$. By induction, superadditivity and submultiplicativity give

$$g(2^n) = 2^n \qquad (n = 0, 1, 2, \ldots). \tag{1}$$

Moreover, $g$ is strictly increasing, since

$$g(y+x) \geq g(y) + g(x) > g(y) \qquad (x, y > 0).$$

**Step 2. Establish an upper bound.**
Fix $t > 0$. For sufficiently large $n$, let

$$m_n = \left\lfloor \frac{2^n}{t} \right\rfloor.$$

Since $m_n t \le 2^n$, monotonicity, superadditivity, and (1) yield

$$2^n = g(2^n) \ge g(m_n t) \ge m_n g(t).$$

As $2^n/m_n \to t$, it follows that

$$g(t) \le t \qquad (t > 0). \tag{2}$$

For every positive integer $k$, we have $g(k) = k$. If $x > 1$, repeated submultiplicativity and monotonicity give

$$g(x)^n \ge g(x^n) \ge g\big(\lfloor x^n \rfloor\big) = \lfloor x^n \rfloor.$$

Taking $n$-th roots and letting $n \to \infty$ gives $g(x) \ge x$. Together with (2), this proves $g(x) = x$ for every $x > 1$.
For $0 < x < 1$, submultiplicativity gives

$$1 = g(1) \le g(x) g\left(\frac{1}{x}\right) = \frac{g(x)}{x},$$

where the last equality follows from the result for $1/x > 1$. Thus $g(x) \ge x$, and (2) forces equality. Therefore $g(x) = x$ for all $x > 0$, and hence $f(x) = 1/x$ for all $x > 0$. □

*Verification.* For $f(x) = 1/x$,

$$f(x+y)\big(f(x) + f(y)\big) = \frac{1}{xy} = f(x)f(y) \le \frac{1}{xy} = f(xy),$$

and $f(2) = 1/2$. Thus the function satisfies all the conditions and is the unique solution. □

## Solution 8. The Multiplicative Mirror

*Proof.* The reciprocal condition gives $f(1) = 1$ and $f(1/x) = 1/f(x)$. The inequalities become

$$\frac{1}{f(x+y)} \ge \frac{1}{f(x)} + \frac{1}{f(y)}, \qquad f(xy)f(x/y) \ge f(x)^2. \tag{1}$$

Replacing $x$ by $1/x$ in the second inequality and using reciprocity gives the reverse inequality. Hence

$$f(xy)f(x/y) = f(x)^2.$$

Applying this to $(u, v)$ and $(v, u)$ and multiplying yields $f(uv)^2 = f(u)^2 f(v)^2$, so positivity gives $f(uv) = f(u)f(v)$.
Set $g = 1/f$. Then $g$ is positive, increasing, superadditive, multiplicative, and $g(1) = 1$.

The reciprocal condition also gives

$$g(1/x) = \frac{1}{f(1/x)} = f(x) = \frac{1}{g(x)}.$$

In particular $g(2) \geq 2$. Moreover, using this identity with $x = f(2)$,

$$g(g(2)) = \frac{1}{f(1/f(2))} = f(f(2)) \leq 2.$$

Since $g(2) \geq 2$ and $g$ is increasing, $2 \geq g(g(2)) \geq g(2) \geq 2$, so $g(2) = 2$.
Multiplicativity gives $g(2^q) = 2^q$ for every rational $q$. These arguments form a dense subset of $(0, \infty)$; monotonicity therefore forces $g(x) = x$ everywhere. Thus $f(x) = 1/x$. Direct substitution shows this function satisfies all conditions with equality. □

## Solution 9. The Reflected Product

*Proof.* **Step 1. Derive multiplicativity.**
The reciprocal identity at $x = 1$ gives $f(1) = 1$. Superadditivity shows that $f$ is strictly increasing: if $v > u > 0$, then

$$f(v) \geq f(u) + f(v - u) > f(u),$$

because $f(v - u) > 0$. In particular,

$$f(2) \geq f(1) + f(1) = 2.$$

Together with the given bound, this gives $f(2) = 2$.
Apply the second inequality with $x$ replaced by $1/x$. Using the reciprocal identity, we obtain

$$\frac{1}{f(x)^2} \leq \frac{1}{f(xy)f(x/y)},$$

so $f(x)^2 \geq f(xy)f(x/y)$. Hence

$$f(x)^2 = f(xy)f(x/y) \qquad (x, y > 0). \tag{1}$$

Taking $y = x$ in (1) gives $f(x^2) = f(x)^2$. Then, for any $a, b > 0$, use (1) with $x = \sqrt{ab}$ and $y = \sqrt{a/b}$ to get

$$f(ab) = f(\sqrt{ab})^2 = f(a)f(b).$$

Thus $f$ is multiplicative.

**Method I. Dyadic squeeze.**
Multiplicativity and $f(2) = 2$ give $f(2^n) = 2^n$ for every integer $n \geq 0$. Fix $t > 0$ and, for sufficiently large $n$, set

$$m_n = \left\lfloor \frac{2^n}{t} \right\rfloor.$$

Since $m_n t \leq 2^n$, monotonicity and repeated superadditivity yield

$$2^n = f(2^n) \geq f(m_n t) \geq m_n f(t).$$

As $2^n/m_n \to t$, it follows that $f(t) \le t$. The reciprocal identity then gives

$$f(t) = \frac{1}{f(1/t)} \ge t.$$

Therefore $f(t) = t$ for all $t > 0$.

**Method II. Dense fixed points.**
For every rational number $q = m/n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, multiplicativity implies

$$f(2^q)^n = f(2^m) = 2^m,$$

where negative integer powers follow from $f(1) = 1$ and multiplicativity. Since $f(2^q) > 0$, we get $f(2^q) = 2^q$. The set $\{2^q : q \in \mathbb{Q}\}$ is dense in $(0, \infty)$; monotonicity and these fixed points imply $f(x) = x$ for every $x > 0$.
Finally, $f(x) = x$ satisfies the reciprocal identity and $f(2) = 2$; both inequalities hold with equality. Hence it is the unique solution. □

## Solution 10. The Reciprocal Pinch

*Proof.* The reciprocal identity gives $f(1) = 1$. By superadditivity,

$$f(2) \ge 2f(1) = 2.$$

Writing $a = f(2)$, we have $a \ge 2$ and $f(1/2) = 1/a$. Thus

$$a + \frac{1}{a} \le \frac{5}{2},$$

or $(2a - 1)(a - 2) \le 0$. Since $a \ge 2$, this forces

$$f(2) = 2. \tag{1}$$

Applying the multiplicative inequality to $1/x, 1/y$ and using the reciprocal identity gives

$$\frac{1}{f(xy)} = f\left(\frac{1}{xy}\right) \ge f\left(\frac{1}{x}\right) f\left(\frac{1}{y}\right) = \frac{1}{f(x)f(y)}.$$

Together with the original inequality, this proves

$$f(xy) = f(x)f(y). \tag{2}$$

Also, $f$ is strictly increasing: if $x < y$, then

$$f(y) = f(x + (y - x)) \ge f(x) + f(y - x) > f(x).$$

By (2) and (1), $f(2^n) = 2^n$ for every integer $n \ge 0$. Fix $t > 0$, and for sufficiently large $n$ put

$$m_n = \left\lfloor \frac{2^n}{t} \right\rfloor.$$

Since $m_n t \le 2^n$, monotonicity and repeated superadditivity give

$$2^n = f(2^n) \ge f(m_n t) \ge m_n f(t).$$

As $2^n/m_n \to t$, we obtain $f(t) \le t$. Applying this bound to $1/t$ and using the reciprocal identity yields

$$f(t) = \frac{1}{f(1/t)} \ge t.$$

Hence $f(t) = t$ for every $t > 0$. □

The identity function directly satisfies all the conditions, so it is the unique solution.

## Solution 11. The Harmonic–Product Squeeze

*Proof.* Define

$$g(t) = \frac{1}{f(1/t)} \qquad (t > 0).$$

Taking reciprocals in the first inequality and substituting $x = 1/u$ and $y = 1/v$ gives

$$g(u + v) \ge g(u) + g(v). \tag{1}$$

The second inequality, applied to $1/u$ and $1/v$ and then reciprocated, gives

$$g(uv) \le g(u)g(v). \tag{2}$$

Thus $g$ is positive, superadditive, and submultiplicative.

**Step 1. Obtain a lower bound for $g$.**
Taking $x = y = 1$ in (2) gives $g(1) \le g(1)^2$, so $g(1) \ge 1$. Also, (1) and positivity show that $g$ is strictly increasing: if $y > x > 0$, then

$$g(y) = g(x + (y - x)) \ge g(x) + g(y - x) > g(x).$$

Fix $t > 1$ and a positive integer $n$. Submultiplicativity and superadditivity give

$$g(t)^n \ge g(t^n) \ge \lfloor t^n \rfloor g(1) \ge \lfloor t^n \rfloor.$$

The middle inequality follows by writing $t^n$ as the sum of $\lfloor t^n \rfloor$ copies of 1 and, if needed, a positive remainder. Taking $n$th roots and letting $n \to \infty$ yields

$$g(t) \ge t \qquad (t > 1). \tag{3}$$

Let $a = 1/c > 1$. By the definition of $g$,

$$g(a) = \frac{1}{f(c)}, \qquad \frac{1}{g(a)} = f(c),$$

and therefore

$$g(g(a)) = \frac{1}{f(1/g(a))} = \frac{1}{f(f(c))} \le \frac{1}{c} = a.$$

On the other hand, (3) applied first to $a$ and then to $g(a) > 1$ gives

$$g(g(a)) \ge g(a) \ge a.$$

Therefore,

$$g(a) = a. \tag{4}$$

**Step 2. Establish a global upper bound.**
By (2) and (4), $g(a^n) \leq g(a)^n = a^n$ for every positive integer $n$. Fix $x > 0$ and, for sufficiently large $n$, put

$$m_n = \left\lfloor \frac{a^n}{x} \right\rfloor \geq 1.$$

Since $m_n x \leq a^n$, monotonicity and repeated superadditivity give

$$a^n \geq g(a^n) \geq g(m_n x) \geq m_n g(x).$$

Consequently,

$$g(x) \leq \frac{a^n}{\lfloor a^n/x \rfloor} \longrightarrow x,$$

so $g(x) \leq x$ for all $x > 0$.

**Step 3. Force equality.**
The upper bound gives $g(1) \leq 1$, and hence $g(1) = 1$. By (2),

$$1 = g(1) = g(x \cdot 1/x) \leq g(x)g(1/x) \leq \frac{g(x)}{x}.$$

Thus $g(x) \geq x$. Together with the upper bound, this proves $g(x) = x$ for all $x > 0$. Returning to the definition of $g$,

$$\frac{1}{f(1/x)} = x,$$

so $f(x) = x$ for every $x > 0$.

**Verification.**
For $f(x) = x$, both inequalities hold with equality, and $f(f(c)) = c$. Hence the identity function is indeed the unique solution. □

## Part III. Iteration, Fixed Points, and Growth

**Solution 12. IMO 1977 – Problem 6**

*Proof.* Start with $a_0 = 2$ and, while $a_j > 1$, set $a_{j+1} = f(a_j - 1)$. If the sequence never reaches 1, then

$$f(a_j) > f(f(a_j - 1)) = f(a_{j+1})$$

produces an infinite decreasing sequence of positive integers. Hence $a_j = 1$ for some $j$, so $f(t) = 1$ for $t = a_{j-1} - 1$. If $t > 1$, the condition at $t-1$ gives $1 = f(t) > f(f(t-1)) \geq 1$, impossible. Thus $t = 1$ and $f(1) = 1$.
Inductively, $f(n) \geq k$ for all $n \geq k$: if it holds for $k$, then for $n \geq k+1$, $f(n-1) \geq k$ and

$$f(n) > f(f(n-1)) \geq k,$$

so $f(n) \geq k + 1$. In particular, $f(n) \geq n$.
It follows that $f(f(n)) \geq f(n)$, hence $f(n+1) > f(n)$; thus $f$ is strictly increasing. If $f(n) > n$, then $f(n) \geq n + 1$, so $f(f(n)) \geq f(n + 1)$, contradicting the hypothesis. Therefore $f(n) = n$ for every positive integer $n$, and direct substitution verifies the solution. □

**Solution 13. The Fixed-Point Lever**

*Proof.* Setting $y = x$ in the given inequality gives

$$f(x)\big(x + f(x)\big) \geq f(f(x))\big(f(x) + x\big).$$

Since $x + f(x) > 0$, it follows that

$$f(f(x)) \leq f(x). \tag{1}$$

**Step 1. Use the assumed two-cycle.**
Let $b = f(a)$. Then $f(b) = a$. Applying (1) at $a$ gives $a \leq b$, while applying it at $b$ gives

$$b = f(f(b)) \leq f(b) = a.$$

Thus $a = b$ and $f(a) = a$.
Now set $x = a$ in the original inequality. Since $f(a) = f(f(a)) = a$, we get

$$a\big(a + f(y)\big) \geq a(a + y),$$

so

$$f(y) \geq y \qquad (y > 0). \tag{2}$$

**Step 2. Force idempotence.**
Applying (2) to $f(x) > 0$ yields $f(f(x)) \geq f(x)$. Together with (1), this gives

$$f(f(x)) = f(x). \tag{3}$$

Substituting (3) into the original inequality and dividing by $f(x) > 0$ yields

$$x + f(y) \geq f(x) + y.$$

Therefore $f(x) - x \leq f(y) - y$ for all $x, y > 0$. Interchanging $x$ and $y$ shows that $f(x) - x$ is a constant $c$. By (2), $c \geq 0$.

Finally, $f(x) = x + c$ and (3) imply

$$x + 2c = f(f(x)) = f(x) = x + c,$$

so $c = 0$. Hence $f(x) = x$ for every $x > 0$.

**Verification.**
For $f(x) = x$, the inequality holds with equality, and $f(f(a)) = a$ for every $a > 0$. □

**Solution 14. IMO 2023 Shortlist – Algebra A4**

*Proof.* Write $f^n(x)$ for the $n$-fold composition of $f$, with $f^0(x) = x$.

**Step 1.** Setting $y = x$ gives

$$2xf(x) \geq (f^2(x) + x)f(x).$$

Since $f(x) > 0$,

$$\boxed{f^2(x) \leq x.} \tag{1}$$

**Step 2.** Setting $x = f(y)$ yields

$$f(y) - f^3(y) \geq y - f^2(y).$$

Replacing $y$ by $f^n(y)$ gives

$$f^{n+1}(y) - f^{n+3}(y) \geq f^n(y) - f^{n+2}(y).$$

Define

$$D_n = f^n(y) - f^{n+2}(y).$$

Then

$$D_{n+1} \geq D_n \geq D_0 = y - f^2(y) \geq 0.$$

For every positive integer $m$,

$$\begin{aligned} y - f^{2m}(y) &= \sum_{k=0}^{m-1} D_{2k} \\ &\geq m\big(y - f^2(y)\big). \end{aligned}$$

Since $f^{2m}(y) > 0$,

$$y > m\big(y - f^2(y)\big).$$

As $m$ is arbitrarily large,

$$\boxed{f^2(y) = y.} \tag{2}$$

**Step 3.** Using (2), the original inequality becomes

$$x(f(x) + f(y)) \geq (x + y)f(y),$$

hence

$$xf(x) \geq yf(y).$$

Interchanging $x$ and $y$ gives equality:

$$xf(x) = yf(y) = c$$

for some $c > 0$. Therefore,

$$\boxed{f(x) = \frac{c}{x}, \qquad c > 0.}$$

Conversely, for $f(x) = c/x$, we have $f(f(x)) = x$ and

$$x(f(x) + f(y)) = c + \frac{cx}{y} = (f(f(x)) + y)f(y).$$

Thus every such function satisfies the given inequality. □

**Solution 15. FEOO 2020 Shortlist – Algebra A1**

*Proof.* Let $P(x, y)$ denote the given inequality.

**Step 1. Determine $f(1)$.**
From $P(1, 1)$,

$$2f(1) \geq 1 + f(1)^2,$$

which gives

$$(f(1) - 1)^2 \leq 0.$$

Thus

$$\boxed{f(1) = 1.} \tag{1}$$

**Step 2. Establish a lower bound.**
By $P(x, 1)$ and (1),

$$2xf(x) \geq x^2 + xf(x).$$

Hence

$$\boxed{xf(x) \geq x^2 \qquad (x \in \mathbb{R}).} \tag{2}$$

**Step 3. Force equality.**
For $x \neq 0$, applying $P(x, 1/x)$ gives

$$2f(1) \geq 1 + xf(x)f\left(\frac{1}{x^2}\right).$$

By (2) at $1/x^2$, we have $f(1/x^2) \geq 1/x^2 > 0$. Thus multiplying $xf(x) \geq x^2$ by this positive value preserves the inequality. Using (1) and (2),

$$\begin{aligned} 1 &\geq xf(x)f\left(\frac{1}{x^2}\right) \\ &\geq x^2 f\left(\frac{1}{x^2}\right) \\ &\geq x^2 \cdot \frac{1}{x^2} = 1. \end{aligned}$$

All inequalities are equalities. In particular,

$$xf(x) = x^2.$$

Therefore,

$$\boxed{f(x) = x \qquad (x \neq 0).} \tag{3}$$

**Step 4. Determine $f(0)$.**
Setting $y = 0$ gives

$$0 \geq xf(x)f(0).$$

For $x \neq 0$, by (3),

$$0 \geq x^2 f(0).$$

Consequently,

$$\boxed{f(0) \leq 0.}$$

**Verification.**
Let

$$f(x) = \begin{cases} x, & x \neq 0, \\ c, & x = 0, \end{cases} \qquad c \leq 0.$$

If $xy \neq 0$, both sides of the original inequality equal $2x^2y^2$.
If $x = 0$, both sides vanish.
If $y = 0$, the inequality becomes

$$0 \geq x^2 c,$$

which holds because $c \leq 0$.
Thus all such functions satisfy the given inequality. □

## Solution 16. The Iterated Ratio Squeeze

*Proof.* For positive $u, v$, define

$$R(u, v) = \frac{u}{v} + \frac{v}{u}.$$

The first inequality in the statement says

$$R(f(x), f(y)) \geq R(x, y).$$

Applying this inequality successively to $(f(x), f(y))$ and $(f^{\circ 2}(x), f^{\circ 2}(y))$ gives

$$R(f^{\circ 3}(x), f^{\circ 3}(y)) \geq R(f^{\circ 2}(x), f^{\circ 2}(y)) \geq R(f(x), f(y)) \geq R(x, y).$$

The second inequality in the statement gives the reverse bound $R(x, y) \geq R(f^{\circ 3}(x), f^{\circ 3}(y))$. Hence equality holds throughout, and in particular

$$R(f(x), f(y)) = R(x, y). \tag{1}$$

**Step 1. Determine the form of $f$.**
Let $c = f(1) > 0$. Taking $y = 1$ in (1),

$$\frac{f(x)}{c} + \frac{c}{f(x)} = x + \frac{1}{x}.$$

For positive $t$, the equation $t + t^{-1} = x + x^{-1}$ has roots $t = x$ and $t = x^{-1}$. Therefore, for every $x > 0$,

$$f(x) = cx \quad \text{or} \quad f(x) = \frac{c}{x}. \tag{2}$$

The choice in (2) cannot differ for two inputs other than 1. Indeed, if $u, v \neq 1$ satisfied $f(u) = cu$ and $f(v) = c/v$, then (1) would give

$$uv + \frac{1}{uv} = \frac{u}{v} + \frac{v}{u},$$

which is equivalent to $(u^2 - 1)(v^2 - 1) = 0$, a contradiction. Thus the same choice holds for every $x \neq 1$. At $x = 1$ the two expressions coincide, so globally either $f(x) = cx$ or $f(x) = c/x$.

**Step 2. Apply the fixed-point condition.**
If $f(x) = cx$, then $f(a) = a$ gives $c = 1$. If $f(x) = c/x$, then $f(a) = a$ gives $c = a^2$. Hence the only possibilities are

$$f(x) = x \qquad \text{or} \qquad f(x) = \frac{a^2}{x}.$$

**Verification.**
Both functions satisfy $f(a) = a$ and preserve $R(x, y)$, so both inequalities hold with equality. Therefore both are solutions. □

## Solution 17. IMO 2022 Shortlist – Algebra A3

*Proof.* Call a pair $(x, y)$ *good* if

$$xf(y) + yf(x) \leq 2.$$

**Step 1. Every good pair has $x = y$.**
Suppose $(x, y)$ is good with $x \neq y$. By symmetry and uniqueness, neither $(x, x)$ nor $(y, y)$ can be good. Thus

$$xf(x) > 1, \qquad yf(y) > 1.$$

By AM–GM,

$$\begin{aligned} xf(y) + yf(x) &\geq 2\sqrt{xf(y) \cdot yf(x)} \\ &= 2\sqrt{xf(x) \cdot yf(y)} \\ &> 2, \end{aligned}$$

a contradiction.
Therefore every good pair satisfies $x = y$.

**Step 2. Establish an upper bound.**
For every $x > 0$, a good pair exists. By Step 1, it must be $(x, x)$. Hence

$$2xf(x) \leq 2,$$

so

$$\boxed{f(x) \leq \frac{1}{x}.} \tag{1}$$

**Step 3. Force equality.**
Fix $x > 0$ and let

$$t = \frac{1}{f(x)} > 0.$$

By (1),

$$f(t) \le \frac{1}{t} = f(x).$$

Therefore,

$$\begin{aligned} xf(t) + tf(x) &\le xf(x) + 1 \\ &\le 2. \end{aligned}$$

Thus $(x, t)$ is a good pair. By Step 1, $t = x$, which gives

$$\boxed{f(x) = \frac{1}{x} \qquad (x > 0).}$$

**Verification.**
For $f(x) = 1/x$, the given inequality becomes

$$\frac{x}{y} + \frac{y}{x} \le 2.$$

However, AM–GM gives

$$\frac{x}{y} + \frac{y}{x} \ge 2,$$

with equality if and only if $x = y$.
Hence, for every $x > 0$, there is exactly one admissible $y > 0$, namely $y = x$.
Therefore $f(x) = 1/x$ is the unique solution. □

**Alternative 1.1. A direct contradiction.**
We use the bound $f(x) \le 1/x$ established above. Suppose that $f(x) < 1/x$ for some $x > 0$. Then the interval $(x, 1/f(x))$ is nonempty. For every $a$ in this interval,

$$af(x) < 1, \qquad xf(a) \le \frac{x}{a} < 1,$$

so

$$af(x) + xf(a) < 2.$$

Thus every such $a$ is a good partner of $x$. Since the interval contains more than one number, this contradicts uniqueness. Hence $f(x) = 1/x$.

**Alternative 1.2. Strict decrease.**
We again use the fact that every good pair is diagonal and $f(x) \le 1/x$. If $y > x$, then $(x, y)$ is not good, whereas $(y, y)$ is good. Therefore

$$xf(y) + yf(x) > 2 \ge 2yf(y) > xf(y) + yf(y),$$

which gives $f(x) > f(y)$. Thus $f$ is strictly decreasing.
For any $y > x$, strict decrease and the fact that $(x, y)$ is not good give

$$2 < xf(y) + yf(x) < (x + y)f(x).$$

Letting $y \to x^+$ yields $f(x) \geq 1/x$. Together with the upper bound, this proves $f(x) = 1/x$.

**Alternative 2. A partner map.**
For each $x > 0$, let $g(x)$ be the unique good partner of $x$. Symmetry of the condition and uniqueness imply $g(g(x)) = x$.
We first show that $f$ is strictly decreasing. If $x < y$, then $(g(y), y)$ is good, but $(g(y), x)$ is not, since $y$ is the unique good partner of $g(y)$. Hence

$$xf(g(y)) + g(y)f(x) > 2 \geq yf(g(y)) + g(y)f(y).$$

Rearranging shows $f(x) > f(y)$.
Now suppose that $(x, y)$ is a good pair with $x \neq y$. Uniqueness implies that neither $(x, x)$ nor $(y, y)$ is good, so $xf(x) + yf(y) > 2$. But $(x, y)$ being good gives

$$(x - y)(f(x) - f(y)) = xf(x) + yf(y) - xf(y) - yf(x) > 0,$$

contradicting strict decrease. Thus every good pair is diagonal, giving $f(x) \leq 1/x$. Alternative 1.1 now forces equality.

**Alternative 3. Equality at the unique partner.**
Let $g(x)$ again denote the unique good partner of $x$. Symmetry gives $g(g(x)) = x$. We claim that every good pair satisfies equality. Suppose instead that $(x, y)$ is good and

$$\delta = 2 - xf(y) - yf(x) > 0.$$

For $0 < \varepsilon < \delta/(2f(x))$, uniqueness makes $(x, y + \varepsilon)$ not good, and therefore

$$f(y + \varepsilon) > \frac{2 - (y + \varepsilon)f(x)}{x} > f(y) + \frac{\delta}{2x}.$$

Put $z = y + \varepsilon$. The good pair $(z, g(z))$ and the fact that $(g(z), y)$ is not good give

$$f(g(z)) \leq \frac{2 - g(z)f(z)}{z}, \qquad f(g(z)) > \frac{2 - g(z)f(y)}{y}.$$

Combining these inequalities yields

$$2\varepsilon < g(z)\big[zf(y) - yf(z)\big] < g(z)\left[\varepsilon f(y) - \frac{y\delta}{2x}\right].$$

For sufficiently small positive $\varepsilon$, the final bracket is negative, a contradiction. Thus every good pair has value exactly 2.
Every pair other than $(x, g(x))$ is not good, so

$$xf(y) + yf(x) \geq 2 \qquad (x, y > 0).$$

Taking $y = x$ gives $f(x) \geq 1/x$. Applying this bound to the good pair $(x, g(x))$ gives

$$2 \geq xf(g(x)) + g(x)f(x) \geq \frac{x}{g(x)} + \frac{g(x)}{x} \geq 2.$$

Equality throughout implies $f(x) = 1/x$.

**Alternative 4. Strict convexity and continuity.**
Let $g(w)$ be the unique good partner of $w$, and define

$$q_s(t) = f(t) + st.$$

For $s = f(g(w))/g(w)$, the good-pair condition gives $q_s(w) \le 2/g(w)$. If $t \neq w$, then $(g(w), t)$ is not good, so $q_s(t) > 2/g(w)$. Thus $q_s$ has a unique global minimum at $w$.
We claim that $f$ is strictly convex. Otherwise, for some $u < v$ and $\lambda \in (0, 1)$, setting $w = \lambda u + (1 - \lambda)v$ would give

$$f(w) \ge \lambda f(u) + (1 - \lambda)f(v).$$

Adding the linear term $st$ yields $q_s(w) \ge \lambda q_s(u) + (1 - \lambda)q_s(v)$, contradicting the strict minimality of $q_s$ at $w$. Hence $f$ is strictly convex and, as a finite convex function on an open interval, is continuous.
As in the first solution, every good pair is diagonal, so $f(x) \le 1/x$. If this inequality were strict at some $x$, the continuous function $h(y) = xf(y)+yf(x)$ would satisfy $h(x) = 2xf(x) < 2$. It would then be less than 2 for all $y$ in some interval around $x$, contradicting the uniqueness of the good partner. Therefore $f(x) = 1/x$ for all $x > 0$.

## Solution 18. The Geometric-Mean Trap

*Proof.* Write $P(x, y)$ for the inequality in the statement. It is symmetric in $x$ and $y$. On the diagonal, $P(x, x)$ is equivalent to

$$\frac{2x}{f(x)} \le \frac{2f(x)}{x},$$

which, since $x, f(x) > 0$, is equivalent to

$$f(x) \ge x. \tag{1}$$

**Step 1. Every admissible pair is diagonal.**
Suppose $P(x, y)$ holds for some $x \neq y$. If $f(x) \ge x$, then $P(x, x)$ also holds, contradicting uniqueness for this fixed $x$. Thus $f(x) < x$. By symmetry, $f(y) < y$ as well. Therefore,

$$\frac{x}{f(y)} + \frac{y}{f(x)} > \frac{x}{y} + \frac{y}{x} \ge 2,$$

whereas

$$\frac{f(x)}{x} + \frac{f(y)}{y} < 2.$$

This contradicts $P(x, y)$. Hence every admissible pair has $x = y$.
For each $x > 0$, an admissible $y$ exists, so it must equal $x$. Consequently, $P(x, x)$ holds for every $x > 0$, and (1) gives

$$f(x) \ge x \qquad (x > 0). \tag{2}$$

**Step 2. Use a geometric-mean choice.**

Suppose $f(a) > a$ for some $a > 0$, and put

$$u = \frac{f(a)}{a} > 1, \qquad y = a\sqrt{u} = \sqrt{af(a)}.$$

By (2), $f(y) \geq y$. Hence

$$\frac{a}{f(y)} + \frac{y}{f(a)} \leq \frac{a}{y} + \frac{y}{f(a)} = \frac{1}{\sqrt{u}} + \frac{1}{\sqrt{u}} = \frac{2}{\sqrt{u}} < 2.$$

On the other hand,

$$\frac{f(a)}{a} + \frac{f(y)}{y} \geq u + 1 > 2.$$

Thus $P(a, y)$ holds. Since $u > 1$, $y \neq a$, contradicting Step 1. Therefore $f(a) \leq a$ for every $a > 0$. Together with (2), this proves $f(x) = x$ for all $x > 0$.

**Verification.**
For $f(x) = x$, the inequality becomes

$$\frac{x}{y} + \frac{y}{x} \leq 2.$$

By AM–GM, the left-hand side is at least 2, with equality if and only if $x = y$. Thus exactly one $y > 0$ works for each $x > 0$, namely $y = x$. □

## Solution 19. The Reciprocal-Slope Pinch

*Proof.* **Step 1. Establish a lower bound.**
Since $f(xf(x)) > 0$, the given inequality implies

$$f(x + y) > y \qquad (x, y > 0).$$

Fix $z > 0$ and take any $x \in (0, z)$ with $y = z - x$. Then $f(z) > z - x$. As this holds for every such $x$, it follows that

$$f(z) \geq z \qquad (z > 0). \tag{1}$$

**Step 2. Control the excess over the identity.**
Define $g(x) = f(x) - x$, so $g(x) \geq 0$ by (1). Since $f(xf(x)) = xf(x) + g(xf(x))$, the original inequality is equivalent to

$$g(x + y) - g(x) \geq \frac{g(xf(x))}{x} \qquad (x, y > 0). \tag{2}$$

In particular, $g$ is nondecreasing.
If $f(1) > 1$, taking $x = 1$ and $y = f(1) - 1$ in the original inequality would give

$$f(1) - 1 + f(f(1)) \leq f(f(1)),$$

which is impossible. Hence $f(1) \leq 1$, and (1) gives $f(1) = 1$; therefore $g(1) = 0$. Since $g$ is nonnegative and nondecreasing,

$$g(x) = 0 \qquad (0 < x \leq 1). \tag{3}$$

**Step 3. Rule out positive excess above 1.**
Suppose $g(a) = c > 0$ for some $a > 1$. For every $u > 1$, we have $uf(u) \geq u^2 > u$. Monotonicity of $g$ and (2) therefore imply, for all $v > u > 1$,

$$g(v) \geq g(u) + \frac{g(uf(u))}{u} \geq g(u) + \frac{g(u)}{u}. \tag{4}$$

Fix $X > a$. For any positive integer $N$, choose

$$u_j = a + \frac{j(X-a)}{N+1} \qquad (0 \leq j \leq N).$$

All these points lie in $[a, X)$, so $g(u_j) \geq c$ and $u_j < X$. Applying (4) to each successive pair gives

$$g(u_{j+1}) \geq g(u_j) + \frac{c}{X} \qquad (0 \leq j < N).$$

Thus $g(u_N) \geq c + Nc/X$. But monotonicity gives $g(u_N) \leq g(X)$, which is a fixed finite value. Choosing $N$ sufficiently large is a contradiction. Hence $g(x) = 0$ for all $x > 1$ as well. Together with (3), this proves $f(x) = x$ for every $x > 0$.
Finally, the identity function satisfies the original inequality with equality throughout. It is therefore the unique solution. □

## Solution 20. The Cross-Dilated Shift

*Proof.* The identity function satisfies the inequality with equality. We prove it is the only solution.

**Step 1. Establish a lower bound.**
Since $f(xf(y)) > 0$, the given inequality implies $f(x+y) > x$ for all $x, y > 0$. Fix $z > 0$. For each $\varepsilon \in (0, z)$, take $x = z - \varepsilon$ and $y = \varepsilon$. Then $f(z) > z - \varepsilon$; hence

$$f(z) \geq z \qquad (z > 0). \tag{1}$$

**Step 2. Bound the excess over the identity.**
Suppose $f(a) > a + 1$ for some $a > 0$, and set

$$x = \frac{a}{f(a) - 1}.$$

Then $0 < x < 1$ and $xf(a) = x + a$. Applying the given inequality at $(x, a)$ gives

$$x^2 + f(x+a) \leq xf(x+a),$$

which is impossible because $f(x+a) > 0$ and $0 < x < 1$. Therefore $f(a) \leq a + 1$ for all $a > 0$.
Define $g(t) = f(t) - t$. By (1),

$$0 \leq g(t) \leq 1 \qquad (t > 0). \tag{2}$$

Substituting $f(t) = t + g(t)$ into the original inequality and cancelling the common terms yields

$$xg(y) + g\big(xf(y)\big) \leq xg(x+y) \qquad (x, y > 0). \tag{3}$$

**Step 3. Force the excess to vanish.**
Since $g$ is nonnegative, (3) implies $g(y) \le g(x+y)$ for all $x, y > 0$, so $g$ is nondecreasing. Taking $x = 1$ in (3), and using $f(y) \ge y$ and the monotonicity of $g$, gives

$$2g(y) \le g(y) + g\big(f(y)\big) \le g(y+1).$$

Inductively, $g(y+n) \ge 2^n g(y)$ for every positive integer $n$. By (2), this forces $g(y) = 0$ for every $y > 0$. Thus $f(y) = y$ on $(0, \infty)$, as claimed. □

## Part IV. Mean and Norm Squeezes

### Solution 21. 2026 IMO Problem 5

*Proof.* We use the arithmetic-orbit approach from the AoPS Wiki. Write $P(x, y)$ for the given chain. Substituting $(f(x), x)$ squeezes its middle term between two copies of $f(x)$, so

$$f(f(x)) = 2f(x) - x.$$

Set $d(x) = f(x) - x$. Then $d(f(x)) = d(x)$ and

$$f^n(x) = x + nd(x) \qquad (n \ge 0),$$

so positivity of all iterates gives $d(x) \ge 0$.
Suppose $A = d(x) > 0$ and $B = d(y) > 0$. For integers $m \ge 0$, $n \ge 1$, set $X = x + mA$ and $Y = y + nB$. Since $d(f(t)) = d(t)$, we have $f(X) = X + A$ and $f(Y - B) = Y$. Applying the right inequality to $(X, Y - B)$ gives

$$\frac{X + Y + A - B}{2} \ge \sqrt{XY}.$$

Squaring and rearranging yields

$$(X - Y)^2 \ge 2(B - A)(X + Y) - (B - A)^2. \tag{1}$$

If $A < B$, then for each sufficiently large $n$ choose $m \ge 0$ so that $|X - Y| \le A/2$; this is possible because the values $x + mA$ have spacing $A$ and $Y$ tends to infinity. The left side of (1) is then at most $A^2/4$, while its right side tends to $+\infty$, a contradiction. Interchanging $x, y$ rules out $B < A$. Hence all positive values of $d$ equal one constant $c > 0$, if any occur.
Thus $d$ takes only the values $0, c$. If $d(x) = c$ and $d(y) = 0$, the left inequality in $P(x, y)$ gives

$$(x - y)^2 \ge 2c(x + y) + c^2 > c^2.$$

So the two level sets are relatively open; connectedness of $(0, \infty)$ forces one to be empty. Therefore $d$ is constant and $f(x) = x + c$ for some $c \ge 0$.
For these functions, the chain reduces to the RMS–AM–GM inequalities for $x$ and $y + c$, so every $c \ge 0$ works. □

## Solution 22. The Hidden Quadratic Function

*Proof.* Put $g(x) = \sqrt{f(x)}$. The given inequalities are equivalent to

$$g(x+y) \geq g(x) + g(y), \tag{1}$$
$$g(xy) \leq g(x)g(y). \tag{2}$$

Since $g > 0$, condition (1) implies that $g$ is strictly increasing. From (2), $g(1) \geq 1$.

**Step 1. Show that $g(2) = 2$.**
By (1),

$$g(2) \geq 2g(1) \geq 2, \qquad g(4) \geq 4g(1) \geq 4.$$

The condition $f(f(2)) \leq 16$ gives

$$g(g(2)^2) \leq 4.$$

Since $g(2)^2 \geq 4$ and $g$ is increasing,

$$4 \geq g(g(2)^2) \geq g(4) \geq 4g(1) \geq 4.$$

Thus $g(1) = 1$ and, by strict monotonicity,

$$\boxed{g(2) = 2.}$$

**Step 2. Show that $g(x) \leq x$.**
For every positive integer $n$, (1) and (2) give

$$2^n = 2^n g(1) \leq g(2^n) \leq g(2)^n = 2^n.$$

Hence

$$\boxed{g(2^n) = 2^n.}$$

Fix $x > 0$ and put

$$m_n = \left\lfloor \frac{2^n}{x} \right\rfloor.$$

For sufficiently large $n$, we have $m_n \geq 1$. By monotonicity and (1),

$$2^n = g(2^n) \geq g(m_n x) \geq m_n g(x).$$

Consequently,

$$g(x) \leq \frac{2^n}{\lfloor 2^n/x \rfloor} \xrightarrow[n\to\infty]{} x.$$

Therefore

$$\boxed{g(x) \leq x \qquad (x > 0).}$$

**Step 3. Establish equality.**
By (2) and the preceding bound,

$$1 = g(1) \leq g(x)g(1/x) \leq \frac{g(x)}{x}.$$

Thus $g(x) \geq x$, and hence

$$g(x) = x \qquad (x > 0).$$

It follows that

$$\boxed{f(x) = x^2 \qquad (x > 0).}$$

Finally, this function satisfies $f(f(2)) = 16$ and

$$\sqrt{f(x+y)} = \sqrt{f(x)} + \sqrt{f(y)} = \sqrt{f(xy)}\left(\frac{1}{\sqrt{f(x)}} + \frac{1}{\sqrt{f(y)}}\right) = x + y.$$

Therefore it is indeed the unique solution. □

## Solution 23. The Euclidean–Multiplicative Squeeze

*Proof.* Define

$$g(t) = f(\sqrt{t})^2 \qquad (t > 0).$$

The first inequality in the statement gives

$$g(x+y) \geq g(x) + g(y). \tag{1}$$

For the second inequality, its right-hand side is

$$f(xy)^2 \frac{f(x)^2 + f(y)^2}{f(x)^2 f(y)^2}.$$

Since $f(x)^2 + f(y)^2 > 0$, cancellation yields

$$f(xy)^2 \leq f(x)^2 f(y)^2.$$

Applying this with $\sqrt{x}$ and $\sqrt{y}$ in place of $x$ and $y$ gives

$$g(xy) \leq g(x)g(y). \tag{2}$$

The composition condition is equivalently

$$g(g(2)) = f(f(\sqrt{2}))^2 \leq 2. \tag{3}$$

**Step 1. Determine $g(1)$ and $g(2)$.**
Taking $x = y = 1$ in (2) gives $g(1) \leq g(1)^2$. Since $g(1) > 0$, we have $g(1) \geq 1$. By repeated use of (1),

$$g(n) \geq ng(1) \geq n \qquad (n \in \mathbb{N}).$$

Fix $t > 1$ and a positive integer $n$. Submultiplicativity and superadditivity imply

$$g(t)^n \geq g(t^n) \geq \lfloor t^n \rfloor.$$

Taking $n$th roots and letting $n \to \infty$ gives $g(t) \geq t$. In particular, $g(2) \geq 2$. Applying this bound again at $g(2) > 1$,

$$2 \geq g(g(2)) \geq g(2) \geq 2,$$

where (3) supplies the first inequality. Thus $g(2) = 2$. Now (2) at $x = y = 1$ gives $g(1) \leq g(1)^2$, while (1) and $g(2) = 2$ give $2 \geq 2g(1)$. Therefore $g(1) = 1$.

**Step 2. Obtain a global upper bound.**
For every positive integer $n$, (1) and (2) give

$$2^n \le g(2^n) \le g(2)^n = 2^n,$$

so $g(2^n) = 2^n$. Fix $x > 0$ and, for sufficiently large $n$, let

$$m_n = \left\lfloor \frac{2^n}{x} \right\rfloor \ge 1.$$

Write $2^n = m_n x + r_n$ with $r_n \ge 0$. If $r_n > 0$, repeated superadditivity and positivity of $g(r_n)$ show that

$$2^n = g(2^n) \ge m_n g(x) + g(r_n) \ge m_n g(x).$$

If $r_n = 0$, repeated superadditivity directly gives the same final bound. Hence

$$g(x) \le \frac{2^n}{\lfloor 2^n/x \rfloor} \longrightarrow x,$$

and so $g(x) \le x$ for every $x > 0$.

**Step 3. Force equality.**
By (2), $g(1) \le g(x)g(1/x)$. Using $g(1) = 1$ and the upper bound just proved,

$$1 \le g(x)g(1/x) \le \frac{g(x)}{x}.$$

Thus $g(x) \ge x$, and consequently $g(x) = x$ for every $x > 0$. Since $g(x) = f(\sqrt{x})^2$ and $f$ is positive, it follows that

$$f(x) = x \qquad (x > 0).$$

**Verification.**
For $f(x) = x$, both inequalities hold with equality, and $f(f(\sqrt{2})) = \sqrt{2}$. Therefore this function is indeed the unique solution. □

## Solution 24. The Quadratic Mean Squeeze

*Proof.* Set $g(t) = f(\sqrt{t})$. The given chain becomes

$$\sqrt{\frac{x^2 + g(y)^2}{2}} \ge \frac{g(x) + y}{2} \ge \sqrt{x g(y)}. \tag{1}$$

Taking $x = g(y)$ forces equality throughout, so

$$g(g(y)) = 2g(y) - y.$$

For $d(x) = g(x) - x$, this gives $d(g(x)) = d(x)$ and $g^{\circ n}(x) = x + nd(x)$. Positivity of every iterate implies $d(x) \ge 0$.
Squaring (1) and factoring the difference of the outer squares yields

$$|d(x) - d(y)|\big(g(x) + y + x + g(y)\big) \le (x - g(y))^2. \tag{2}$$

Indeed, each outer square differs from $(x+g(y))^2$ by $(x-g(y))^2$.
Any two positive values of $d$ must coincide. Otherwise, let $d(p)=r>0$, $d(q)=s>0$, $r\neq s$. The iterates $X_n=p+nr$ and $Y_m=q+ms$ retain these respective $d$-values. For each large $n$, choose $m\geq 0$ with $|X_n-Y_m|\leq s/2$. In (2) at $(X_n,Y_m)$, the left side tends to infinity, while $(X_n-g(Y_m))^2=(X_n-Y_m-s)^2\leq 9s^2/4$, a contradiction.
Thus $d$ takes at most two values, 0 and some $c>0$. If both occur, (2) shows that any points with different values are more than $c$ apart. Bisecting an interval whose endpoints have different values produces such pairs arbitrarily close together, impossible. Hence $d$ is constant, say $c\geq 0$, and $f(x)=g(x^2)=x^2+c$.
The composition condition gives

$$a^4\geq f(f(a))=(a^2+c)^2+c,$$

forcing $c=0$. Thus $f(x)=x^2$. Substitution reduces the chain to RMS–AM–GM for $x^2,y^2$, and $f(f(a))=a^4$. □

## Part V. Real-Line and Polynomial Inequalities

### Solution 25. Vietnam Mathematical Olympiad 1991

*Proof.* **Step 1. Determine $f(0)$.**
Setting $x=y=z=0$ gives

$$f(0)-f(0)^2\geq\frac{1}{4},$$

or equivalently,

$$\left(f(0)-\frac{1}{2}\right)^2\leq 0.$$

Hence

$$\boxed{f(0)=\frac{1}{2}.} \tag{1}$$

**Step 2. Establish a key inequality.**
Setting $z=0$ and using (1), we obtain

$$\frac{f(xy)}{2}+\frac{1}{4}-\frac{f(x)}{2}\geq\frac{1}{4}.$$

Therefore,

$$\boxed{f(xy)\geq f(x)}\qquad(x,y\in\mathbb{R}). \tag{2}$$

**Step 3. Prove that $f$ is constant.**
For any $t\neq 0$, taking $x=1,y=t$ in (2) gives

$$f(t)\geq f(1).$$

Taking $x=t,y=1/t$ gives

$$f(1)\geq f(t).$$

Thus

$$f(t)=f(1)\qquad(t\neq 0). \tag{3}$$

Finally, setting $x = y = z = 1$ gives

$$f(1) - f(1)^2 \geq \frac{1}{4},$$

so

$$\left(f(1) - \frac{1}{2}\right)^2 \leq 0.$$

Consequently, $f(1) = \frac{1}{2}$.
Combining this with (1) and (3), we get

$$\boxed{f(x) = \frac{1}{2} \qquad (x \in \mathbb{R}).}$$

Conversely, for $f(x) = \frac{1}{2}$, we have

$$\frac{f(xy) + f(xz)}{2} - f(x)f(yz) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Thus the inequality holds with equality for all real $x, y, z$. □

**Solution 26. IMO 2023 Shortlist – Algebra A2**

*Proof.* **Step 1. Introduce a symmetric substitution.**
Put

$$s = x + y, \qquad t = x - y,$$

so that

$$x = \frac{s+t}{2}, \qquad y = \frac{s-t}{2}.$$

The given inequality becomes

$$f(s)f(t) \geq f\left(\frac{s+t}{2}\right)^2 - f\left(\frac{s-t}{2}\right)^2. \tag{1}$$

**Step 2. Force a sign constraint.**
Replacing $t$ by $-t$ in (1), we obtain

$$f(s)f(-t) \geq f\left(\frac{s-t}{2}\right)^2 - f\left(\frac{s+t}{2}\right)^2. \tag{2}$$

Adding (1) and (2), the squared terms cancel:

$$\boxed{f(s)\big(f(t) + f(-t)\big) \geq 0} \tag{3}$$

for every $s, t \in \mathbb{R}$.

**Step 3. Apply the strictness assumption.**
By assumption, the original inequality is strict for some pair $(x_0, y_0)$.
Set

$$s_0 = x_0 + y_0, \qquad t_0 = x_0 - y_0.$$

Then (1) is strict at $(s_0, t_0)$. Adding the corresponding inequality (2) yields

$$f(s_0)\big(f(t_0) + f(-t_0)\big) > 0.$$

Therefore,

$$\boxed{f(t_0) + f(-t_0) \neq 0.}$$

Now fix $t = t_0$ in (3). For every $s \in \mathbb{R}$,

$$f(s)\big(f(t_0) + f(-t_0)\big) \geq 0.$$

Since the second factor is a fixed nonzero real number, either

$$f(s) \geq 0 \qquad \text{for all } s \in \mathbb{R},$$

or

$$f(s) \leq 0 \qquad \text{for all } s \in \mathbb{R}.$$

This proves the required conclusion. □

## Solution 27. IMO 2011 – Problem 3

*Proof.* Replacing $y$ by $t - x$ in the given inequality gives

$$f(t) \leq (t - x)f(x) + f(f(x)) \qquad (x, t \in \mathbb{R}). \tag{1}$$

**Step 1. Derive a two-variable inequality.**
Set $t = f(z)$ in (1), then add the resulting inequality to the one obtained by interchanging $x$ and $z$. The terms involving $f(f(x))$ and $f(f(z))$ cancel, yielding

$$xf(x) + zf(z) \leq 2f(x)f(z) \qquad (x, z \in \mathbb{R}). \tag{2}$$

Taking $z = 2f(x)$ in (2) cancels the terms involving $f(2f(x))$ and gives

$$xf(x) \leq 0 \qquad (x \in \mathbb{R}). \tag{3}$$

In particular, $f(x) \leq 0$ for $x > 0$, while $f(x) \geq 0$ for $x < 0$.

**Step 2. Show that $f$ is nonpositive everywhere.**
Suppose $f(w) > 0$ for some $w \in \mathbb{R}$. Taking $x = w$ in (1),

$$f(t) \leq (t - w)f(w) + f(f(w)).$$

Choose

$$t < \min\left(0,\, w - \frac{f(f(w))}{f(w)}\right).$$

Then $t < 0$ and the right-hand side is negative, so $f(t) < 0$. This contradicts $f(t) \geq 0$ for $t < 0$, established in Step 1. Therefore $f(x) \leq 0$ for every real $x$.

**Step 3. Prove the required vanishing.**
For $x < 0$, Step 1 gives $f(x) \geq 0$, while Step 2 gives $f(x) \leq 0$. Hence $f(x) = 0$ for all $x < 0$. Using (1) with any $x < 0$ and $t < 0$ now gives

$$0 = f(t) \leq (t - x)f(x) + f(f(x)) = f(0),$$

so $f(0) \geq 0$. Step 2 also gives $f(0) \leq 0$, and therefore $f(0) = 0$. Thus $f(x) = 0$ for every $x \leq 0$. □

## Solution 28. The Degree-Normalized Cubic Squeeze

*Proof.* Define $g(x) = f(x)/x^3$, so $f(x) = x^3 g(x)$. Substitution into the given inequalities gives

$$xg(x) + yg(y) \le (x+y)g(x+y), \qquad g(x) + g(y) \ge 2g(x+y).$$

Equivalently,

$$\frac{xg(x) + yg(y)}{x+y} \le g(x+y) \le \frac{g(x)+g(y)}{2} \qquad (x, y > 0). \tag{1}$$

**Step 1. Establish dyadic invariance and monotonicity.**
Taking $y = x$ in (1) yields

$$g(x) \le g(2x) \le g(x),$$

so

$$g(2x) = g(x) \qquad (x > 0). \tag{2}$$

The lower and upper bounds in (1) also imply

$$2\big(xg(x) + yg(y)\big) \le (x+y)\big(g(x) + g(y)\big),$$

which simplifies to

$$(x-y)\big(g(x) - g(y)\big) \le 0.$$

Thus $g$ is non-increasing.

**Step 2. Show that $g$ is constant.**
Fix $a > 0$. For any $x \in [a, 2a]$, monotonicity and (2) give

$$g(a) = g(2a) \le g(x) \le g(a),$$

so $g(x) = g(a)$ throughout $[a, 2a]$. These intervals cover $(0, \infty)$ as $a$ varies, and therefore $g$ is constant: there is some $c \in \mathbb{R}$ such that $g(x) = c$ for all $x > 0$. Consequently,

$$f(x) = cx^3 \qquad (x > 0).$$

**Verification.**
For any real constant $c$, substituting $f(x) = cx^3$ into the first inequality reduces it to $c(x+y) \le c(x+y)$, and the second to $2c \ge 2c$. Hence every such function satisfies both conditions. □

## Solution 29. The Conditioned Cubic Squeeze

*Proof.* **Step 1. Establish oddness and monotonicity.**
On the plane $a + b + c = 0$, both conditions apply, so

$$f(a^3) + f(b^3) + f(c^3) = 3f(abc). \tag{1}$$

Taking $(a, b, c) = (t, -t, 0)$ and using $f(0) = 0$ gives $f(-t^3) = -f(t^3)$. Since every real number is a cube, $f$ is odd.
If $x \ge y$, take $a = \sqrt[3]{x}$, $b = -\sqrt[3]{y}$, and $c = 0$. Their sum is nonnegative, and the first

condition gives

$$f(x) + f(-y) + f(0) \geq 3f(0).$$

Oddness and $f(0) = 0$ imply $f(x) \geq f(y)$. Thus $f$ is non-decreasing.

**Step 2. Derive a boundary identity.**
In (1), take $(a, b, c) = (u, v, -u - v)$. Oddness yields

$$f((u+v)^3) = f(u^3) + f(v^3) + 3f(uv(u+v)). \tag{2}$$

Expand $f((a+b+c)^3)$ by applying (2) first to $(a+b, c)$ and then to $(a, b)$. Expanding instead first with $(a+c, b)$ gives, after cancelling the common terms,

$$f(ab(a+b)) + f((a+b)c(a+b+c)) = f(ac(a+c)) + f((a+c)b(a+b+c)). \tag{3}$$

**Step 3. Obtain Jensen's equation and affinity.**
Let $x, y > 0$ with $x \neq y$, and set

$$k = \sqrt{\frac{y}{x}} \neq 1, \qquad 2k(k^2-1)t^3 = x,$$

where the nonzero real $t$ exists uniquely. Choose

$$a = (k+1)t, \qquad b = (k-1)t, \qquad c = k(k-1)t.$$

Direct calculation gives

$$ab(a+b) = x, \qquad (a+b)c(a+b+c) = y,$$

and

$$ac(a+c) = (a+c)b(a+b+c) = \frac{x+y}{2}.$$

Substitution in (3) proves

$$f(x) + f(y) = 2f\left(\frac{x+y}{2}\right). \tag{4}$$

This also holds when $x = y$, so (4) holds for all positive $x, y$.
Because $f$ is non-decreasing, the midpoint identity implies that $f$ is affine on $(0, \infty)$. Indeed, repeated midpoint interpolation first gives linear interpolation at dyadic fractions of any interval; monotonicity then squeezes the value at every intermediate point between such dyadic approximations. Thus there are constants $m, b$ such that $f(x) = mx + b$ for all $x > 0$.
Substitute this form into (2), for positive $u, v$. The left side is $m(u+v)^3 + b$, while the right side is $m(u^3 + v^3 + 3uv(u+v)) + 5b = m(u+v)^3 + 5b$. Hence $b = 0$. Since $f(1) = 1$, we get $m = 1$, so $f(x) = x$ for $x > 0$. Oddness and $f(0) = 0$ extend this to every real $x$.

**Verification.**
For the identity function,

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\big((a-b)^2 + (b-c)^2 + (c-a)^2\big).$$

Its sign agrees with the sign of $a + b + c$, so both required inequalities hold. □

## Solution 30. The Quadratic–Difference Squeeze

*Proof.* Taking $y = x$ in the first inequality gives $f(x) \geq x^2$ for every real $x$. Define

$$g(x) = f(x) - x^2.$$

Then $g(x) \geq 0$. Substituting $f(t) = t^2 + g(t)$ into the second inequality and cancelling the quadratic terms yields

$$g(x+y) + g(x-y) \leq 2g(x) \qquad (x, y \in \mathbb{R}). \tag{1}$$

**Step 1. Show that $g$ is constant.**
Fix arbitrary $a, b \in \mathbb{R}$, and set

$$x_k = a + k(b-a) \qquad (k \in \mathbb{Z}), \qquad \Delta_k = g(x_{k+1}) - g(x_k).$$

Applying (1) with $x = x_k$ and $y = b - a$ gives

$$g(x_{k+1}) + g(x_{k-1}) \leq 2g(x_k),$$

so $\Delta_k \leq \Delta_{k-1}$. In particular, $\Delta_j \leq \Delta_0 = g(b) - g(a)$ for every integer $j \geq 0$. For any positive integer $n$, summing these bounds gives

$$-g(a) \leq g(x_n) - g(a) = \sum_{j=0}^{n-1} \Delta_j \leq n\big(g(b) - g(a)\big),$$

where the first inequality uses $g(x_n) \geq 0$. If $g(b) < g(a)$, choosing $n$ sufficiently large makes the rightmost quantity less than $-g(a)$, a contradiction. Thus $g(b) \geq g(a)$. Since $a, b$ were arbitrary, interchanging them shows $g(a) = g(b)$; therefore $g$ is constant. Write $g(x) = d$. Its nonnegativity gives $d \geq 0$, and hence

$$f(x) = x^2 + d \qquad (x \in \mathbb{R}).$$

**Verification.**
For $d \geq 0$,

$$f(x) + f(y) - 2xy = (x-y)^2 + 2d \geq 0,$$

and

$$f(x+y) + f(x-y) = (x+y)^2 + (x-y)^2 + 2d = 2f(x) + 2y^2.$$

Thus every function in the stated family satisfies both inequalities. □

## Solution 31. The Iterated Parallelogram–Quadratic Squeeze

*Proof.* **Step 1. Derive a quadratic lower bound.**
Comparing the outer expressions in the given chain gives

$$f(x) - x^2 + f(f(x)) - f(y) + 2y^2 - 2yf(x) \geq 0.$$

Setting $y = f(x)$ cancels the terms involving $f(f(x))$ and the quadratic terms in $y$,

leaving

$$f(x) \geq x^2 \qquad (x \in \mathbb{R}). \tag{1}$$

Define $g(x) = f(x) - x^2$. Then $g(x) \geq 0$ for every real $x$.

**Step 2. Obtain midpoint concavity of the residual.**
The right-hand inequality and the identity

$$\frac{(x+y)^2 + (x-y)^2}{2} = x^2 + y^2$$

show that

$$g(x+y) + g(x-y) \leq 2g(x) \qquad (x, y \in \mathbb{R}). \tag{2}$$

**Step 3. Prove that $g$ is constant.**
Fix $a, b \in \mathbb{R}$, and write

$$x_k = a + k(b-a) \quad (k \geq 0), \qquad \Delta_k = g(x_{k+1}) - g(x_k).$$

For each $k \geq 1$, apply (2) with $x = x_k$ and $y = b - a$. This gives

$$g(x_{k+1}) + g(x_{k-1}) \leq 2g(x_k),$$

so $\Delta_k \leq \Delta_{k-1}$. Consequently, $\Delta_k \leq \Delta_0 = g(b) - g(a)$ for all $k \geq 0$. For every positive integer $n$, nonnegativity of $g$ now yields

$$-g(a) \leq g(x_n) - g(a) = \sum_{k=0}^{n-1} \Delta_k \leq n\big(g(b) - g(a)\big).$$

Thus $g(b) - g(a) \geq -g(a)/n$ for every $n$, and letting $n \to \infty$ shows $g(b) \geq g(a)$. Since $a, b$ were arbitrary, interchanging them proves $g(a) = g(b)$. Therefore $g(x) = d$ for a constant $d \geq 0$, and

$$f(x) = x^2 + d. \tag{3}$$

**Step 4. Use the composition condition and verify.**
By (3),

$$f(f(2)) = (4+d)^2 + d = 16 + 9d + d^2 \leq 16.$$

Since $d \geq 0$, this forces $d = 0$. Hence $f(x) = x^2$ for every real $x$.
For this function, $f(f(2)) = 16$, the middle and right expressions in the given chain are both $x^2 + y^2$, and the left expression equals

$$x^2 + y^2 - (y - x^2)^2 \leq x^2 + y^2.$$

Thus $f(x) = x^2$ satisfies all the conditions, completing the proof. □

## Solution 32. The Skew-Symmetric Parallelogram Squeeze

*Proof.* **Step 1. Obtain a quadratic lower bound.**

Comparing the outer expressions in the given chain gives

$$2xy + x^2 f(y) - y^2 f(x) \le f(x) + y^2.$$

Setting $y = x$ cancels the skew-symmetric terms, so

$$2x^2 \le f(x) + x^2, \qquad \text{hence} \qquad f(x) \ge x^2 \quad (x \in \mathbb{R}). \tag{1}$$

Define $g(x) = f(x) - x^2$. Then $g(x) \ge 0$ for all real $x$.

**Step 2. Establish midpoint concavity of the residual.**
The right-hand inequality and the parallelogram identity give

$$\frac{(x+y)^2 + (x-y)^2}{2} + \frac{g(x+y) + g(x-y)}{2} \le x^2 + y^2 + g(x).$$

After cancellation,

$$g(x+y) + g(x-y) \le 2g(x) \qquad (x, y \in \mathbb{R}). \tag{2}$$

**Step 3. Prove that $g$ is constant.**
Fix $a, b \in \mathbb{R}$, and put

$$x_k = a + k(b-a) \quad (k \ge 0), \qquad \Delta_k = g(x_{k+1}) - g(x_k).$$

For each $k \ge 1$, applying (2) with $x = x_k$ and $y = b - a$ yields

$$g(x_{k+1}) + g(x_{k-1}) \le 2g(x_k),$$

so $\Delta_k \le \Delta_{k-1}$. Thus $\Delta_k \le \Delta_0 = g(b) - g(a)$ for all $k \ge 0$. For every positive integer $n$, nonnegativity of $g$ implies

$$-g(a) \le g(x_n) - g(a) = \sum_{k=0}^{n-1} \Delta_k \le n\big(g(b) - g(a)\big).$$

Letting $n \to \infty$ gives $g(b) \ge g(a)$. Interchanging $a$ and $b$ shows $g(a) = g(b)$, so $g(x) = d$ for some constant $d \ge 0$. Therefore

$$f(x) = x^2 + d. \tag{3}$$

**Step 4. Eliminate the constant shift.**
Substituting (3) into the outer inequality gives

$$(x-y)^2 + d(1 - x^2 + y^2) \ge 0 \qquad (x, y \in \mathbb{R}).$$

Taking $y = x - 1$ reduces this to

$$1 + d(2 - 2x) \ge 0 \qquad (x \in \mathbb{R}).$$

If $d > 0$, this fails for sufficiently large $x$. Hence $d = 0$ and $f(x) = x^2$ for all real $x$.
For $f(x) = x^2$, the middle and right-hand expressions are both $x^2 + y^2$, while the left-hand expression is $2xy \le x^2 + y^2$. Thus this function satisfies the stated inequalities and is the unique solution. □

## Solution 33. The Dyadic Parallelogram–Quadratic Squeeze

*Proof.* **Step 1. Obtain an inequality for the residual.**
Put $g(x) = f(x) - x^2$. Comparing the outer expressions in the given chain and substituting $f(x) = x^2 + g(x)$ gives

$$2g(x) + 3g(y) + 2(x - 2y)^2 \geq 0 \qquad (x, y \in \mathbb{R}). \tag{1}$$

Taking $x = y = 0$ shows $g(0) \geq 0$, while setting $x = 2t$ and $y = t$ gives

$$2g(2t) + 3g(t) \geq 0 \qquad (t \in \mathbb{R}). \tag{2}$$

**Step 2. Prove that $g$ is nonnegative.**
The right-hand inequality in the problem and the parallelogram identity imply

$$g(x + y) + g(x - y) \leq 2g(x) \qquad (x, y \in \mathbb{R}). \tag{3}$$

Taking $x = y = t$ gives $g(2t) \leq 2g(t) - g(0)$. Combining this with (2),

$$0 \leq 2g(2t) + 3g(t) \leq 7g(t) - 2g(0).$$

Since $g(0) \geq 0$, it follows that $g(t) \geq \frac{2}{7}g(0) \geq 0$ for all $t \in \mathbb{R}$.

**Step 3. Show that $g$ is constant.**
Fix $a, b \in \mathbb{R}$, and define

$$x_k = a + k(b - a) \quad (k \geq 0), \qquad \Delta_k = g(x_{k+1}) - g(x_k).$$

For each $k \geq 1$, apply (3) with $x = x_k$ and $y = b - a$. Then

$$g(x_{k+1}) + g(x_{k-1}) \leq 2g(x_k),$$

so $\Delta_k \leq \Delta_{k-1}$. Therefore $\Delta_k \leq \Delta_0 = g(b) - g(a)$ for all $k \geq 0$. For any positive integer $n$, nonnegativity of $g$ yields

$$-g(a) \leq g(x_n) - g(a) = \sum_{k=0}^{n-1} \Delta_k \leq n\big(g(b) - g(a)\big).$$

Letting $n \to \infty$ gives $g(b) \geq g(a)$. Interchanging $a$ and $b$ shows that $g$ is constant, say $g(x) = d$ with $d \geq 0$. Hence

$$f(x) = x^2 + d. \tag{4}$$

**Step 4. Determine $d$ and verify the solution.**
Using (4) in the composition condition gives

$$1 \geq f(f(1)) = (1 + d)^2 + d = 1 + 3d + d^2.$$

As $d \geq 0$, this forces $d = 0$, so $f(x) = x^2$ for all real $x$.
For $f(x) = x^2$, the middle and right-hand expressions in the problem both equal $5x^2 + 5y^2$, while the left-hand expression is

$$3x^2 + 8xy - 3y^2 = 5x^2 + 5y^2 - 2(x - 2y)^2 \leq 5x^2 + 5y^2.$$

Also $f(f(1)) = 1$. Thus $f(x) = x^2$ satisfies all conditions and is the unique solution. □

# Conclusion

The central lesson of these problems is to keep experimenting with the inequalities: choose useful inputs, look for quantities to compare, and follow a promising bound until it becomes decisive. We hope this booklet encourages you to approach unfamiliar problems with patience, curiosity, and confidence.